

Electoral Accountability and Industrial Change: Regulator Selection and Renewable Energy Growth

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Why Regulators

- ⊙ Representatives delegate policy to regulatory commissions, often because they lack the technical capacity themselves
- ⊙ With rapid industrial and technological change, regulator decisions are likely increasingly important
- ⊙ Theory and empirical evidence indicate that regulator behavior can determine important industry outcomes, which varies based on how they are selected
- ⊙ In the US, regulators are either *appointed* (most common) or *elected* directly by voters (a non-trivial number of states)

Regulator Incentives

- ◉ Selection gives regulators incentives to make systematically different decisions
- ◉ Evidence this shapes industry outcomes (Besley and Coate 2003, Leaver 2009, Lim and Yurukoglu 2018)
- ◉ Besley and Coate (2003) predict and show empirically that direct election → more consumer-oriented policy

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Broadly: Do elections produce 'good' outcomes, i.e. social-welfare-maximizing policies, in a setting that more accurately characterizes modern economies?

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I focus on: the US Electricity Sector 1990–2024

- ◉ Pressure to integrate energy sources beyond what the incumbent fossil fuel industry produces
- ◉ New actors whose preferences cut across the existing interest group structure: firms with a financial interest in new energy sources (renewable energy firms), and groups concerned with the incumbent industry's externality (pro-renewable NGOs)

Besley and Coate (2003)

Two types of citizens with respect to regulatory issues

Consumers (Majority)

No financial stake in the regulated industry

→ Always prefer lower prices

Business Interests (Minority)

Owners of capital, or workers who capture some rents

→ Prefer higher prices

Appointed → Pro-Business: regulation is bundled with other issues in electing the appointer; costs diffuse, benefits concentrated

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Mechanism: selection changes the *signal* a regulator must satisfy to keep the job, determining their objective function

A simple world: two interests and a single axis of conflict, over price

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→ Creates new, cross-cutting coalitions

→ Regulators no longer face pro-business incentives when appointed and pro-consumer incentives when elected

Economics literature: railroads used truck-weight rules against a substitute technology (Stigler 1971); regulatory bargains erode when technology shifts the coalition beneath them (Peltzman 1976, 1989).

New Interest Group Preference Structure

Consumers

- ◉ **Pro-Green consumers:** Consumer blocks who prefer industry change over lowest possible prices due to associated externalities
In theory, this group could be important for electoral outcomes. In practice, voters explicitly willing to pay more for renewables are a small subset, even in the most Democratic parts of the US
- ◉ **Climate NGOs and environmental groups:** Organized non-profit groups who advocate for industry change for social purposes
Membership-based and 'grassroots', but staffed by highly educated professionals with elite access (Skocpol 2003). Can act through campaigns, or inside the regulatory and appointment process

Business Interests

- ◉ **Renewable Energy Producing Firms:** Substitutes for firms in the existing business interest group, against **fossil fuel producers and naturally monopolistic utilities** that have long dominated the regulatory arena

Cross-Cutting Coalitions: Electricity

	Status quo low price today	Transition invest now, save later
Business	Fossil fuel producers; incumbent utilities	Renewable developers; in-state manufacturers
Consumers	Price-sensitive consumers (the large majority)	Green NGOs; pro-transition citizens (small, organized)

- ◉ **Business:** developers need resource plans that retire fossil capacity, rate treatment for transition investment, and siting and interconnection rulings incumbents oppose; utilities vary with asset ownership
- ◉ **Consumers:** a second type with a non-financial stake, willing to pay more today for a faster shift, organized in advocacy groups

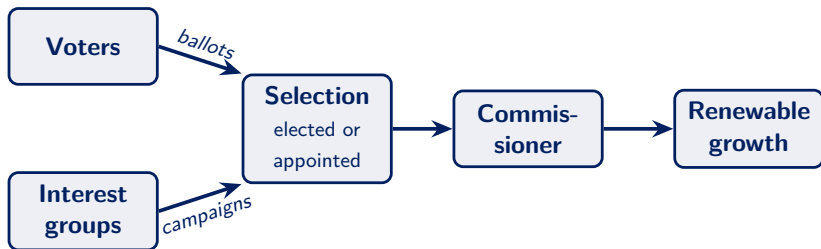
What Is 'Pro-Consumer' Under Industrial Change?

In Besley and Coate, pro-consumer means **low prices today**. Within the energy transition:

- ◉ New sources need investment now, passed to ratepayers in short term, even where renewables are cheaper
- ◉ Fossil fuels limited and costly; alternatives increasingly cheaper
- ◉ Uncontained climate costs fall on *all* consumers
- ◉ Regulatory consequences unfold over horizons far longer than electoral cycles

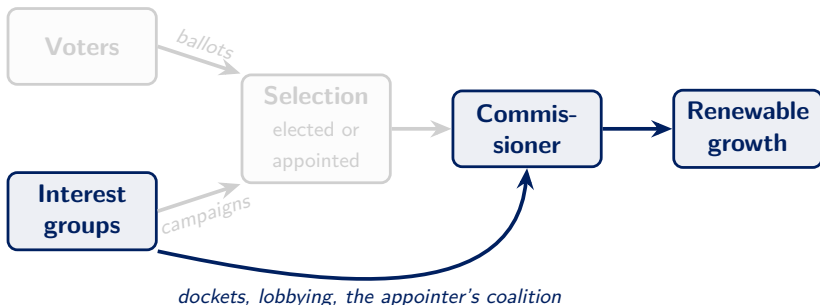
→ What is good for consumers immediately may differ from long run.
Most discount the future so prefer the low price today

My Argument



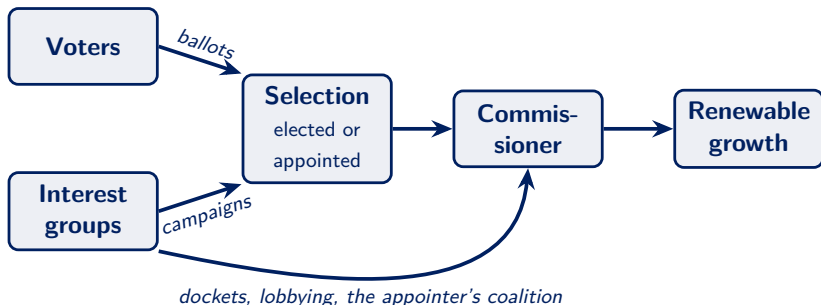
Electoral path. Elected Regulators objective function: appease voters, usually based on bills; incumbent industry interest very well organized and dominate campaign settings

My Argument



Regulatory path. Insulated Regulators: can rely on expertise, testimony, regulatory lobbying. Green groups are weak in elections but strong *inside*: elite orgs, persistence in dockets, and often have a seat in the governor's coalition. Appointment gives them a chance to compete

My Argument



⇒ **Renewable growth slower on average when voters elect their Commission**, though partisanship matters.

► Argument in words

Why Elections May Underinvest in Climate Policy

Even with rational voters and a decisive median voter, elections likely fall short of max social welfare

1. **Externality:** Voter pays the full cost of a higher bill but captures little of the benefit; the social cost of carbon is not internalized, which is why economists assign it to policymakers (Stern 2007; Nordhaus 2007)

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3. **Timing:** Costs now, benefits in future. Evidence of steep, inconsistent discounting (Laibson 1997; Jacobs and Matthews 2012); money-saving energy investments left untaken (Jaffe and Stavins 1994; Allcott and Greenstone 2012)

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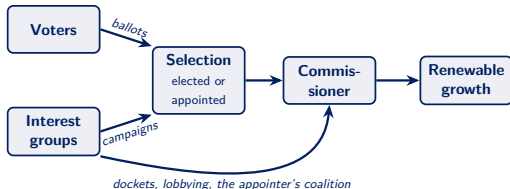
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→ Beyond energy: infrastructure, disaster preparedness (Jacobs 2011; Healy and Malhotra 2009)

In theory, voters underinvest. How much do they actually behave this way?

▶ Classic debates

When Should Selection Matter Most?



Important Variation

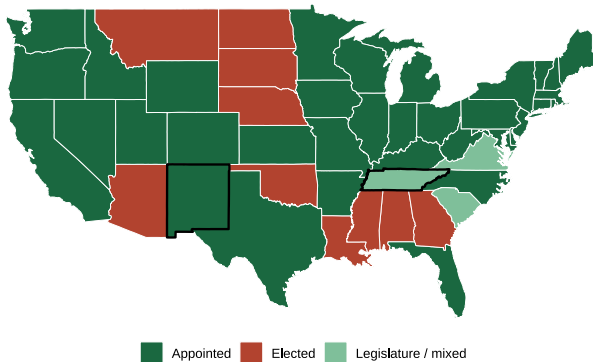
- ◉ Partisanship of state government, and of the Governor (appointer) in particular
- ◉ Alternative energy generation costliness: natural resource endowments or lack thereof
- ◉ Underlying preference distributions
- ◉ Commission authority over generation: vertically integrated vs. restructured; TVA (TN) and public power (NE) outside PUC reach

Setting: Public Utility Commissions (PUCs)

- ◉ Most state PUCs were created in the early 1900s to regulate utilities in the absence of market competition: cost containment, affordable rates
- ◉ Mandate: rates that are 'reasonable and nondiscriminatory', and service of 'good quality'
- ◉ Importance: oversee about 1% of US GDP
- ◉ Today far more than price setting: utility resource plans, rate treatment of new investment, siting and interconnection, grid reliability, distributed energy resources
- ◉ Growing demand (now with AI), cheaper wind and solar, and climate change all push toward integrating alternatives

Scope condition: selection should matter where the Commission actually governs generation (vertically integrated states; investor-owned utilities under PUC jurisdiction), and be weak or absent where restructuring or public ownership removed generation from its reach

How States Select Their Commissioners



- About a dozen states elect; the rest appoint (governor, legislature, or mixed)
- Only two switch, 1990–2024: **TN** (elected → appointed, 1996); **NM** (appointed → elected, 1999; back to appointed, 2023)

Empirical Expectations

- E1.** Elected commissions are associated with less renewable generation on average than appointed commissions
- E2.** Voter support for renewable energy is low, even amongst Democrats, when it carries a direct, short-term cost
- E3.** Green interests have more influence under appointment because they perform better inside the beltway than in campaigns, relative to fossil fuel interests
 - a. They allocate more money and time to inside activity than to campaigns
 - b. Under appointment, their inside resources approach those of fossil fuel firms and utilities; their electoral resources do not
 - c. They look like classic corporate interest groups inside: elite, expertise-based lobbying
 - d. For a given level of effort, their requests fare better before appointed Commissions
- E4.** The elected–appointed gap is largest when the principal that selects the Commission is Democratic: Democratic governors' coalitions include green interests and renewable developers; elected Commissions answer to voters, even when those voters elect Democrats

Research Design

Exploit state-level variation in institutional design, US, 1990–2024

1. **Selection and renewable generation** (E1, E4)

Renewable share of generation, 50 states; original roster of every Commissioner and their partisanship; elected vs. appointed Commissions of the same partisan composition, net of geography, economy, and the electorate's partisanship

2. **Voter behavior** (E2)

Every statewide energy ballot measure 2000–2024; every local renewable or climate measure that imposes a direct cost, where complete lists exist; precinct returns; cast vote records linking each vote to the voter's partisan choices (Kuriwaki et al.)

3. **Interest group allocation** (E3)

Campaign finance in Commissioner elections and energy ballot campaigns; federal lobbying and campaign finance 2000–2022; participation in state PUC proceedings, with effort converted to dollars using state labor costs; matched elected and appointed states; New Mexico before and after 2023; NM Amendment 1 (2020)

Data

- ◉ **Outcome:** share of generation from wind, solar, geothermal, and biomass; 50 states, 1990–2024 (EIA)
 - ◉ Excludes hydro: a legacy of dam building, fixed by geography, that predates every Commissioner [► With hydro](#)
- ◉ **Commissions:** selection method from constitutions and statutes; original roster of every Commissioner
- ◉ **Electorate:** Snyder (2026) normal vote across all statewide contests; governor's party; legislative seats
- ◉ **Controls:** log GDP, manufacturing share, log population (lagged); NREL wind and solar endowment; RPS and restructuring ledgers
- ◉ **Voters:** energy ballot measures, precinct returns, cast vote records
- ◉ **Interest groups:** Follow the Money, OpenSecrets, PUC dockets, IRS 990s

Measuring Commission Partisanship

A new roster: 1,287 Commissioner spells, all 50 states, 1990–2026 (name, seat, dates, selecting authority), reconciled against statutory seat schedules

Problem: appointed Commissioners have no official party

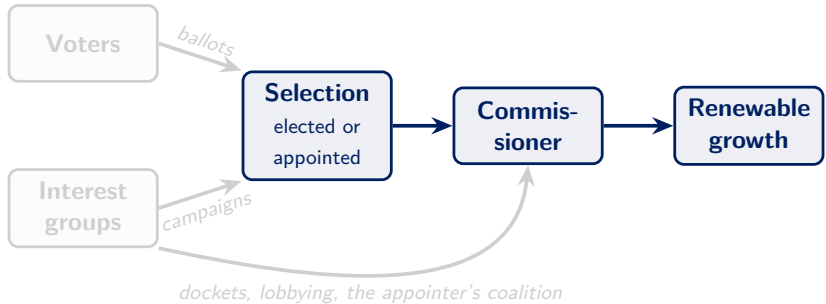
- ◉ *Appointer rule* (primary): party of the governor at appointment, or of the selecting legislature or body
- ◉ Staggered terms outlast appointers: a Commission is a layered artifact of past appointments
- ◉ *PUC Democratic Composition*: Democratic share sitting on July 1; 1,848 of 1,850 state-years

Validation: MSU Institute of Public Utilities archive (personal party, 2000–2020)

- ◉ Agreement: 65 of 67 elected (97%); 426 of 595 appointed (72%)
- ◉ Personal-party benchmark (1,464 state-years) used as a check

► Roster

Commissioner Selection and Renewable Generation



At the same partisan composition, do elected Commissions produce less renewable growth?

The Identifying Variation

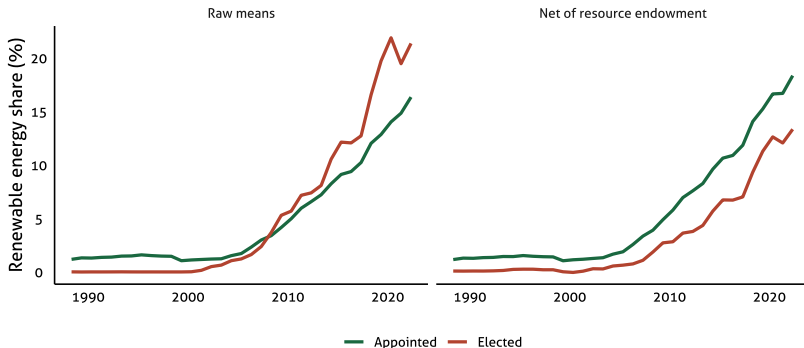
- ◉ Commission composition varies within nearly every state: 48 of 50 states; within-state SD 0.34 (overall 0.40)
- ◉ The selection institution changes in only **two**: Tennessee (1996) and New Mexico (1999, 2023)
- ◉ → With state FE, the elected coefficient comes from TN and NM alone: the estimates show how TN and NM changed when their method changed

Specification (state s , year t ; institutions and controls lagged one year)

$$\text{Renew}_{st} = \beta_1 \text{Elected}_{s,t-1} + \beta_2 \text{DemComp}_{s,t-1} + \beta_3 \text{Elected}_{s,t-1} \times \text{DemComp}_{s,t-1} \\ + X'_{s,t-1} \gamma + \alpha_{s \text{ or } r} + \delta_t [+ \lambda_s t] + \varepsilon_{st}$$

- ◉ Quantity of interest: elected minus appointed at a 50–50 Commission, $\beta_1 + 0.5 \beta_3$
- ◉ SEs clustered by state, plus inference built for few switching states

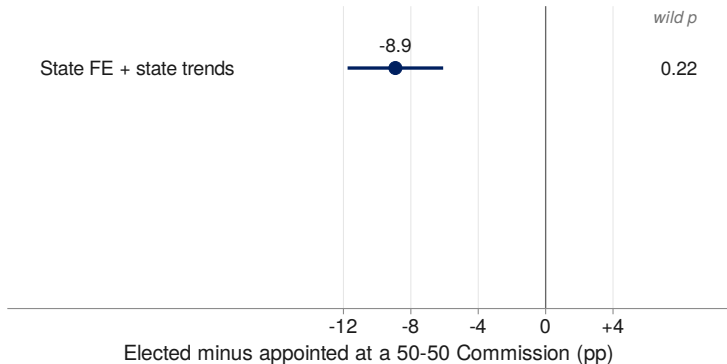
Natural Resources



Mean renewable share by system: raw (left); net of NREL wind and solar endowment within year (right).

- In raw means, elected states generate *more*
- Six of twelve elected states are in the Plains wind belt (ND, SD, OK, NE, NM, MT): about double the developable wind of appointed states, which include the low-resource Southeast

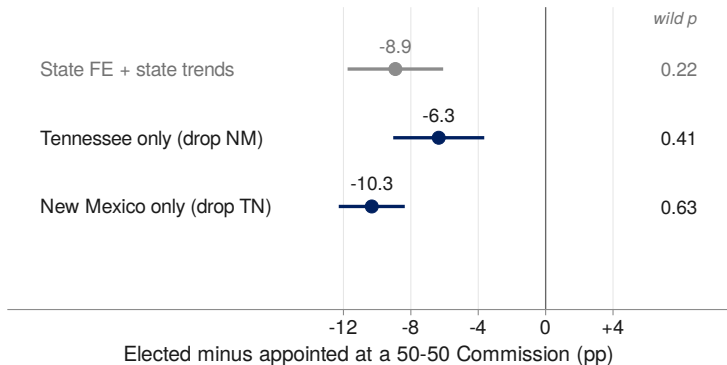
Comparisons Within States



State FE and state-specific trends: while elected, renewable share **8.9 pp** lower (at a 50–50 Commission)

State FE absorb each state's endowment; state trends absorb the growing payoff to it (the Plains wind build-out, which coincides with NM's elected period). Two switching states: wild-bootstrap p at right. [► Why two states](#) [► Authority](#) [► Full table](#)

Comparisons Within States



Neither state drives it alone: Tennessee only **-6.3**; New Mexico only **-10.3**

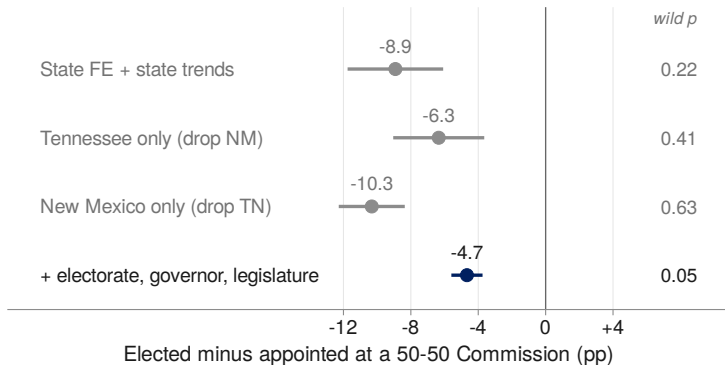
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Comparisons Within States



Adding the electorate's partisanship, the governor's party, and the legislature: **-4.7**

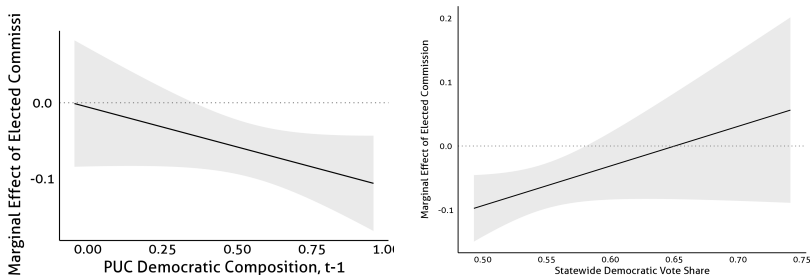
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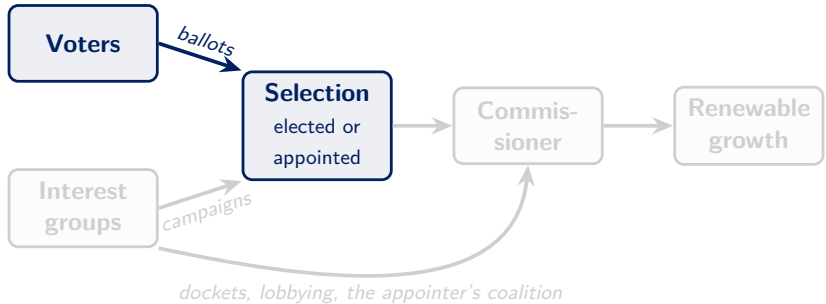
[► Full table](#)

Is the Gap Concentrated Among Democratic Commissions?



Marginal effect of an elected Commission, Democratic vote share ≥ 0.5 . Left: by composition (personal party). Right: by electorate's Democratic share.

- Personal party (left): the gap widens as the Commission becomes more Democratic
- Appointer rule, within state: roughly flat (-11.1 all-R, -7.5 all-D)
- By electorate (right): narrows as more Dem but indistinguishable from 0 for most about 55



Can Voters be Driving an Electoral Gap?

Voters and Climate Policy: What We Know

Mostly stated support: broad support that fades with cost and depends on design and framing (Ansolabehere and Konisky 2014; Stokes and Warshaw 2017)

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Revealed support: single measures, disasters, parties

- California environmental initiatives track income, ideology, partisanship (Kahn and Matsusaka 1997; Wu and Cutter 2011)
- Ideology explains most support for Washington's carbon tax (Anderson, Marinescu, and Shor 2023)
- Wildfires raise pro-environment voting among Democrats only (Hazlett and Mildenberger 2020); disasters and Swiss referendums (Baccini and Leemann 2021)
- Voters punish incumbents for local wind siting (Stokes 2016)
- Green-party support is different: a party vote imposes no direct cost, and support for a party varies and can also strategically motivation.

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Do voters, including Democrats, support renewable energy? Particularly *when it costs them*?

Full review: companion paper on the universe of energy and climate ballot measures.

Testing the Voter Assumption

- I assume consumers prioritize low prices over renewables
- If voters reliably endorse renewable related policies, elected Commissions face pressure *toward* the transition
- Surveys likely over-estimate support.
- What do voters want when renewables may carry a short-term, direct cost?

Behavioral Measure of Voter Support for Renewable Energy

- Statewide: every energy measure 2000–2024;
- Local: Subset for initial effort for 2000–2026, Ballotpedia and states with repositories. Focus on cost-imposing measures vs. free advisory questions

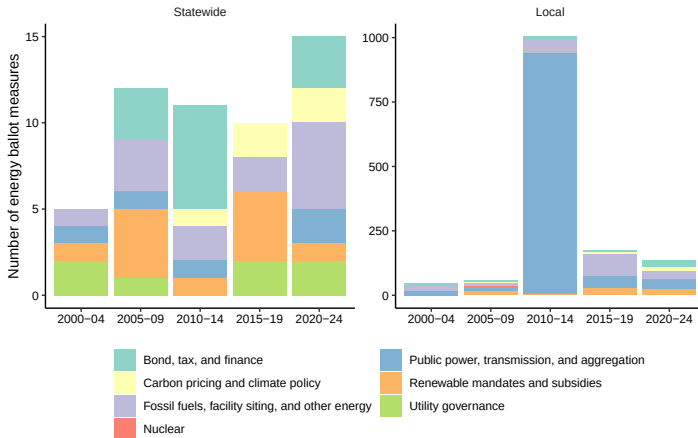
Analytical Unit:

- Precinct Aggregates
- Individual ballots: Cast Vote Records (Baltz et al. 2024)

Energy on the Ballot

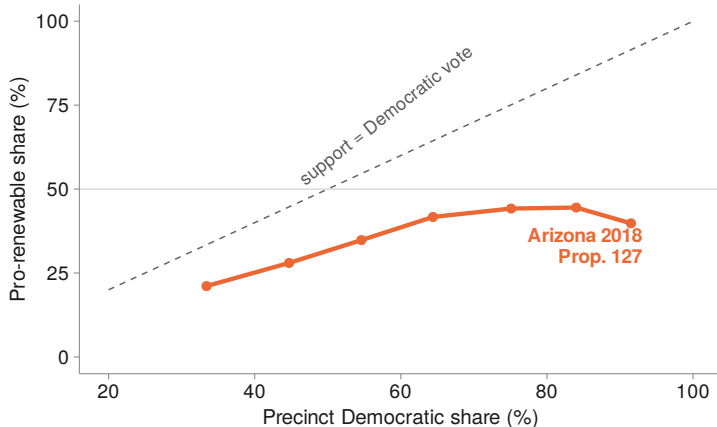
Measure type	Statewide	Local	Total
Fossil fuels, facility siting, and other energy	13	204	217
Bond, tax, and finance	12	70	82
Renewable mandates and subsidies	11	64	75
Utility governance	7	0	7
Public power, transmission, and aggregation	5	1,056	1,061
Carbon pricing and climate policy	5	23	28
Nuclear	0	5	5
All energy measures	53	1,422	1,475

Energy on the Ballot, 2000–2024



- Mostly referral, bond, and governance routes, not citizen-initiated
- 53 statewide measures; **26** on renewables or climate directly
- Pro-renewable side won 15, but **lost every vote to price carbon**

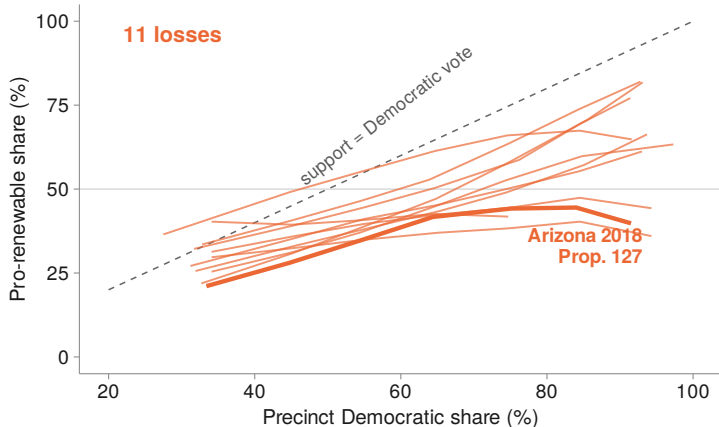
Support Trails Democratic Partisanship



Arizona 2018, 50% renewable mandate: support rises with partisanship, far below the Democratic vote, and never reaches a majority

Vote-weighted means in 10-point bins of the Democratic share of the top-of-ticket race; 22 votes with precinct or town returns.

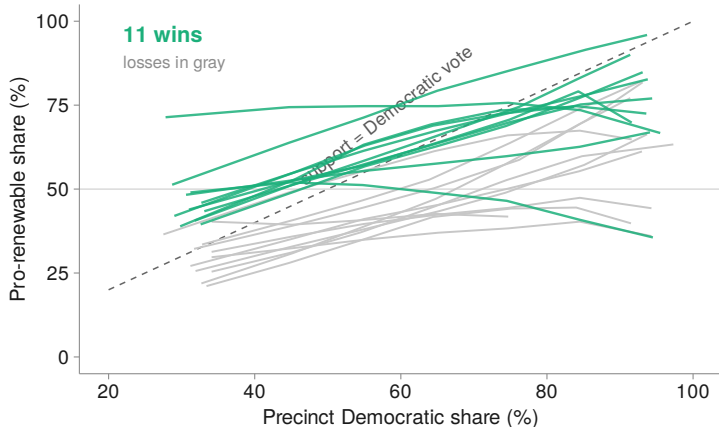
Support Trails Democratic Partisanship



All 11 losses: in precincts $\geq 70\%$ Democratic, the measures trailed the Democratic vote by **27 points** (median)

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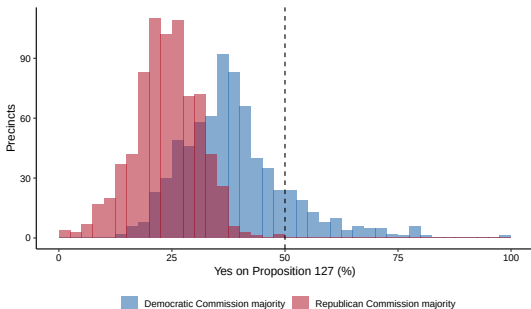
Support Trails Democratic Partisanship



The 11 wins run closer to the Democratic vote, but still rise less than one for one [► By measure](#)

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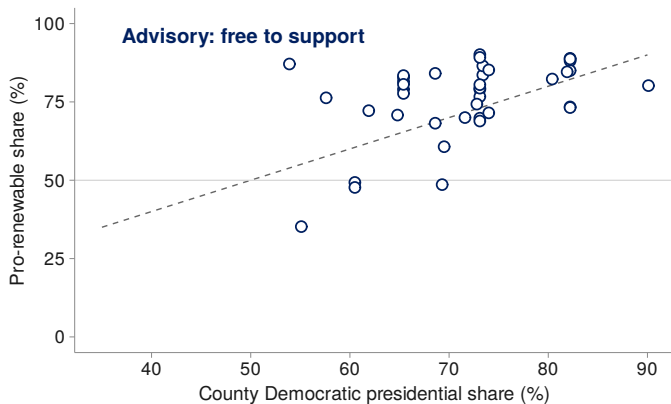
Arizona 2018: Candidate vs. Policy on the Same Ballot



Precinct support for Prop. 127 (50% RPS), by whether precinct was Dem Commission majority.

- Only election with Commissioners and a renewable mandate on one ballot
- Democratic candidates (one ran on renewables) carried half of precincts; Kennedy won. Prop. 127 lost 31–69
- **85%** of Democratic-candidate-majority precincts voted **No**; where Yes won, the two Dems ran within ~ 2 pts: a party vote, not an issue vote

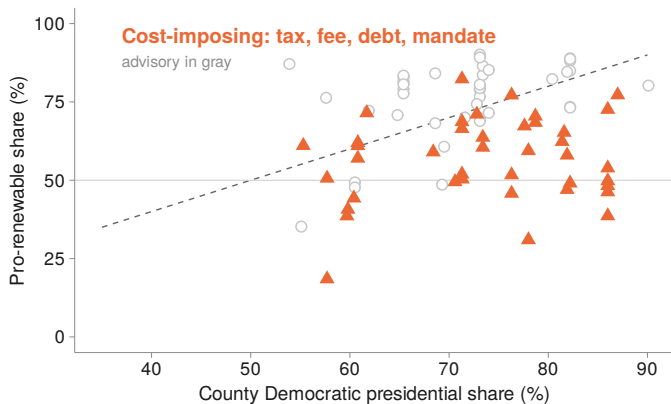
Local Measures: Free Support vs. Costly Support



Advisory questions (85; 47 with returns): **69%** on average, **80%** where Democrats won $\geq 70\%$

Complete lists where they exist (Ballotpedia; official IL, CA, MA lists), frame fixed before outcomes.

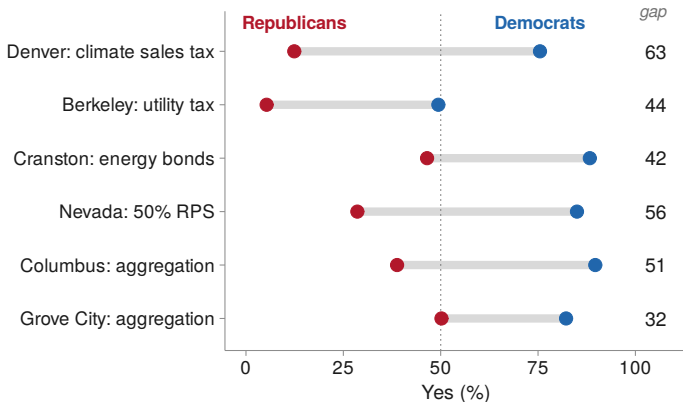
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Cost-imposing measures (53; 41 with returns): **57%**. Where Democrats won $\geq 70\%$, they lost 10 of 34; within Berkeley and SF, **32 pts** behind advisory [► Full table](#)

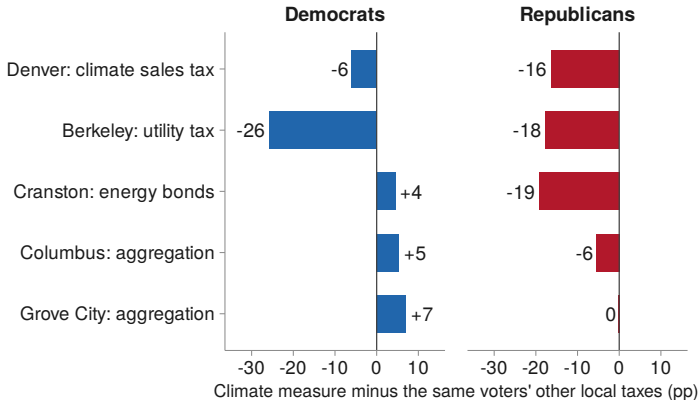
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Individual Voters: 2020 Cast Vote Records



Party of each voter's U.S. House vote. Party gaps of 42–63 pts on the costly measures, vs. 19 pts on 2020 local taxes nationally (Dem. 71.8%, Rep. 52.4%)

Individual Voters: 2020 Cast Vote Records



Same voter, same ballot: Republicans discount costly climate measures by **16–19 pts**, by ≤ 6 when there is no new cost; Berkeley's Democrats by 26

[► Full](#)

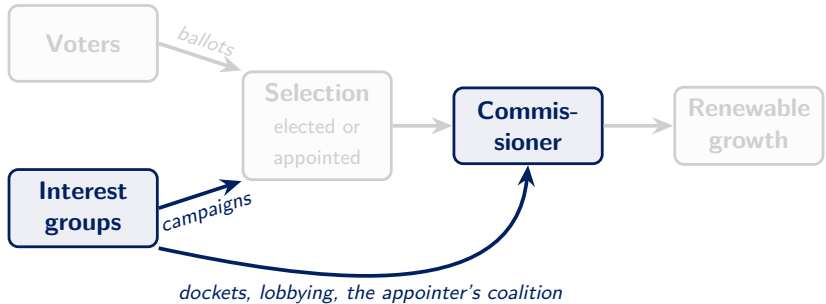
Voters: What We Learn

- ⊙ Renewable mandates rarely reach the ballot; when they fail, support trails Democratic partisanship by 11–43 pts in strongly Democratic precincts
- ⊙ Electing a regulator identified with renewables does not mean supporting the policy
- ⊙ Local measures that impose costs draw ~ 20 pts less support than free advisory questions in the most Democratic places
- ⊙ Republicans abandon their usual support for local taxes when a measure carries a climate cost, so these measures pass or fail on Democratic votes alone

→ An elected Commissioner, however Democratic, faces a re-election signal that penalizes acting on the transition

→ The transition's electoral coalition is not the Democratic electorate; it is the **most Democratic fraction** of it

Interest Group Effort Allocation



Green Interest Group Activity: Tips Balance?

Evidence of Interest Group Preference and Behavior

Pro-Alternative Energy Groups (Consumer Interests): least is known about their 'campaign' and 'lobbying' activity

E3a More resources to regulatory participation than to elections

E3b Closer to incumbents' resources inside than in the electoral arena

E3d Given recorded effort, requests fare better before appointed Commissions

Evidence

1. **Federal record** 2000–2022: both channels, same organizations
2. **State campaigns:** Commissioner races, energy ballot committees
3. **PUC dockets**, first 180 days of every eligible proceeding (IRPs, rate cases, renewable programs, fuel cost, certificates)
 - ⦿ Matched elected–appointed pairs: AZ–NV, GA–KY, LA–AR, MS–WV
 - ⦿ New Mexico before and after its 2023 switch to appointment

Measuring Effort Inside the Beltway

Each docket record: organization and family, reach, proceeding type, document type, pages, date; each request records issue, action, position, and disposition in the final order

Preparation-cost equivalent of document d of type k :

$$C_d = r_s(a_k + b_k P_d)$$

- ◉ P_d prepared pages; a_k fixed preparation time; b_k hours per page
- ◉ r_s : \$300/hour (2023 dollars, from intervenor-compensation materials), or scaled by the state's lawyer wage (BLS OEWS)
- ◉ Benchmarked against organizational budgets (IRS Form 990)

Filers: green NGOs, renewable firms, utilities, fossil fuel firms, other non-profits; public advocates separate; counsel is not the client; joint filings attributed once

State Campaigns: Green Money Is Marginal

One high-profile energy referendum vs. 25 years of Commissioner elections in the archetypal elected state

Context	Total spending	Green nonprofit \$
Maine Q1 Yes side (Block NECEC)	\$27.0M	\$284k
Maine Q1 No side (Allow NECEC)	\$67.4M	\$0
<i>ME Q1 Total</i>	<i>\$107M</i>	<i>\$284k (0.27%)</i>
GA PSC Democratic candidates, 1998–2022	\$4.3M	\$49k
GA PSC Republican candidates, 1998–2022	\$5.5M	\$61k
<i>GA PSC Total</i>	<i>\$9.9M</i>	<i>\$110k (1.1%)</i>

- One ballot measure drew $\sim 10\times$ all GA PSC races over 25 years
- Corporate donors: 99%+ of Maine Q1, 78–81% of GA PSC. Maine's 'Yes': NextEra (\$20.3M), Vistra, Calpine, protecting market share
- Sierra Club Georgia: \$3,500 to PSC candidates in 25 years; energy-sector money 91% to Republican incumbents, solar firms included

Follow the Money. GA PSC: 13,830 contribution rows, 65 candidates. ME Q1: 1,622 rows, 8 committees.

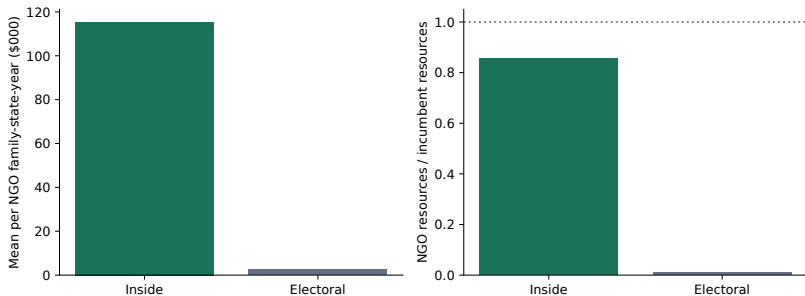
Georgia PSC: Green Money Does Not Follow Positions

Hand-coded renewable positions of all 65 GA PSC candidacies, 1998–2022, linked to every contribution

Position on renewables	N	Raised (\$M)	Pro-env. (\$k)	Alt.-energy firms (\$k)	Green share (%)
Strongly pro-renewable	9	2.68	10.9	4.6	0.6
Lean pro-renewable	18	2.12	7.0	22.4	1.4
Mixed or conflicting	4	0.62	5.0	24.0	4.7
No identifiable position	30	2.85	9.9	6.2	0.6
Lean or strongly anti	4	1.63	6.6	8.0	0.9
All candidacies	65	9.90	39.5	65.2	1.1

- ◉ Nearly half take no position; zero LCV endorsements, three Sierra Club
- ◉ Strongly pro-renewable candidates: same green share as those with no position
- ◉ Sierra Club-endorsed Lindy Miller raised \$1.7M (2018), \$201 of it from pro-environmental donors

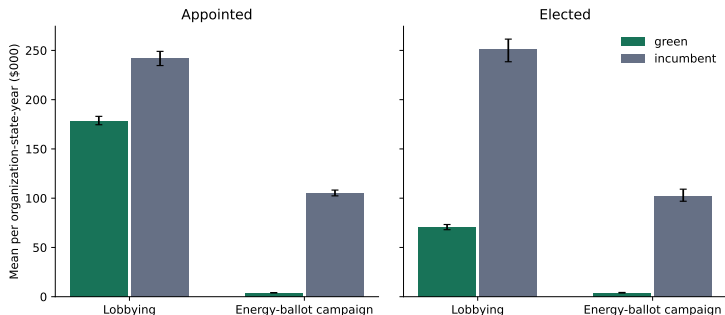
Allocation Across Channels



Left: mean NGO resources per family–state–year. Right: relative to utility and fossil-firm resources, same sample.

- ◉ Same family, state, year; electoral side inclusive (candidates, IEs, energy ballot measures) where an election existed
- ◉ NGO preparation-cost equivalents: \$115,393 vs. \$2,545 electoral
- ◉ Relative to incumbents: **0.86** inside vs. **0.010** electorally
- ◉ NGO time: 642 preparation hours vs. 20 campaign hours

Resources by Selection System



Resource allocation 2017–2024, per organization–state–year. Intervals resample states.

Green / incumbent mean resources	Lobbying	Campaign
Appointed systems	0.74	0.04
Elected systems	0.28	0.04

Under appointment, green lobbying approaches incumbents'; campaign resources are a small fraction in both

Participation and Dispositions in the Matched Pairs

Denominator: *all* eligible proceedings, including those with no green intervenor

Appointed minus elected	Estimate	95% interval
Participated in eligible proceeding	0.114	[0.095, 0.132]
Substantive filings	3.39	[3.09, 3.69]
Log preparation effort	2.05	[1.86, 2.24]
Full grant, conditional on effort	0.245	[0.180, 0.310]
Full or partial grant, conditional on effort	0.271	[0.229, 0.312]

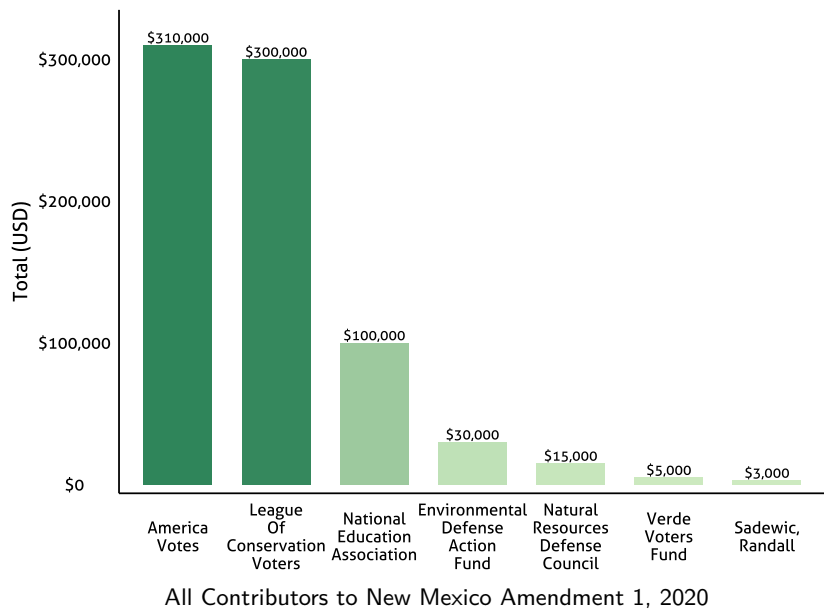
- ⊙ Pair, year, proceeding controls; dispositions add issue, position, prior filings and preparation cost; state-clustered CR1
- ⊙ Conditioning separates the *return* to effort from effort itself
- ⊙ Green filings look professional: 73% substantive, 27% testimony (utilities 62%, 22%)
- ⊙ Limits: 8 states; recorded work is not all work

Evidence of Interest Group Preference and Behavior

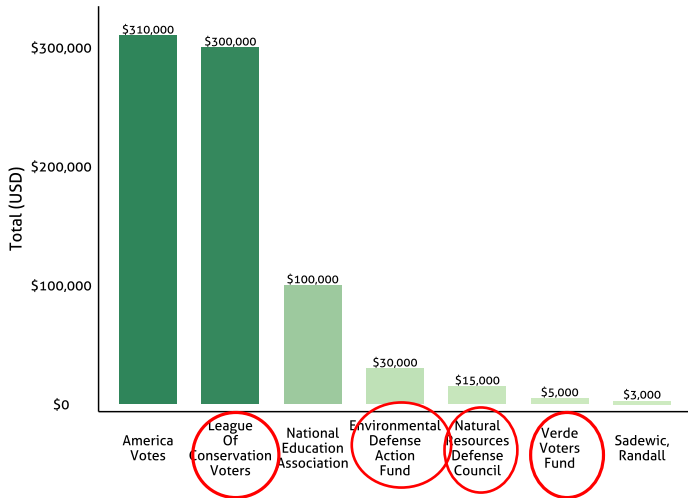
Pro-Alternative Energy Groups (Consumer Interests)

- ◉ New Mexico Amendment 1, 2020
- ◉ Voted to change their PUC from an elected five-person Commission to an appointment-based three-person Commission; passed with 51%
- ◉ Same mechanism created the five-person elected Commission in 1996
- ◉ How did the measure end up on the ballot? Who supported it?
- ◉ All funding in favor came from six non-profit organizations, four of them climate and alternative energy focused

Evidence of Interest Group Preference and Behavior

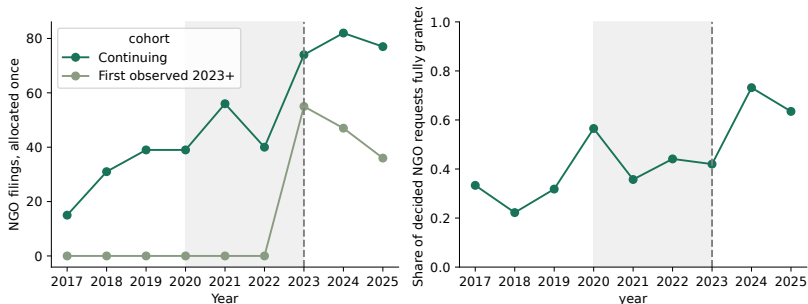


Evidence of Interest Group Preference and Behavior



LCV alone gave \$300,000: $\sim 85\times$ Sierra Club Georgia's \$3,500 to PSC candidates over 25 years. Groups that decline to fund Commissioner candidates pay to abolish Commissioner elections

New Mexico: Before and After Appointment



NM NGO activity and request dispositions. Shading: 2020–2022; dashed line: 2023. Panels have different denominators.

- The one state moving from election to appointment in the window
- Continuing organizations (intensify?) vs. first observed after 2023 (entry?)
- 2020–22 anticipation kept separate; utilities a comparison, not a placebo

Interest Groups: What We Learn

- ◉ Where both channels are observed, environmental organizations tilt overwhelmingly toward lobbying (36:1), and are more competitive with industry inside than in elections
- ◉ In Commissioner elections, green money is marginal ($\leq 1.1\%$ of spending) and does not follow pro-renewable candidates
- ◉ Inside regulatory proceedings, green organizations do professional, expertise-based work and come much closer to parity with incumbents, especially under appointment
- ◉ Revealed preference: when offered the chance to abolish Commissioner elections, they paid for it

→ Despite their grassroots nature, green NGOs behave much like classic corporate interest groups

Summary

Question: Under industrial change, does electoral accountability yield regulators who promote a long-term energy transition?

- ◉ **Selection and generation:** in the two switching states, renewables grew more slowly while Commissions were elected (suggestive)
- ◉ **Voters:** little electoral pressure toward the transition. Mandates trail Democratic partisanship; costly local measures draw far less support than free ones; Republicans withdraw their usual support for local taxes when a climate cost is attached
- ◉ **Interest groups:** green organizations are far more productive at lobbying than at electoral campaigning, which gives them more influence when regulators are insulated from voters

→ Elections may not produce the best long-term outcomes for consumers when outcomes involve complex temporal trade-offs

Implications and Contributions

- ◉ **Regulator selection:** extends Besley and Coate (2003) to a world where industrial change splits both coalitions. Elected regulators may still be 'pro-consumer' in the narrow sense of prices today
- ◉ **Scope:** appointment is not universally better for the transition. Appointed Commissions follow the governor, so a Republican governor in a blue state produces a Commission as Republican as a red state's. Appointment is where green interests have their best *chance* to compete
- ◉ **Civil society:** consumer advocacy groups shift away from citizens toward insulated channels of influence. Rational adaptation, or a departure from their democratic mandate?
- ◉ **Climate opinion:** builds on ballot-measure studies (Kahn and Matsusaka 1997; Anderson et al. 2023) with behavioral evidence on *costly vs. free* support, within parties and on individual ballots
- ◉ **Beyond energy:** any policy with immediate costs and delayed, diffuse benefits should face the same electoral headwind (Jacobs 2011; Healy and Malhotra 2009)

Next Steps

- ◉ Outcomes Commissions decide directly: resource plans, rate treatment, interconnection rulings, where selection's effect on decisions is observed rather than inferred from generation
- ◉ Complete the collection of PUC docket records for the matched pairs and New Mexico, and a second coder for request dispositions
- ◉ RPS stringency (required renewable share each year) in place of a binary indicator
- ◉ Companion paper: the full universe of energy and climate ballot measures since 2000, state by state, with capture–recapture estimates of what remains unobserved
- ◉ More cast vote records: California counties with card-level records, Los Angeles

Thank you

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Appendix

- A. Selection and generation
- B. Voters
- C. Interest groups
- D. Anticipated questions
- E. The argument in detail

A. Selection and Generation

Summary Statistics

	Mean	SD	Min	Max	N
Renewable energy share	0.050	0.090	0.000	0.641	1,750
Renewable and hydro energy share	0.155	0.214	0.000	0.943	1,750
RPS active	0.366	0.482	0.000	1.000	1,750
PUC Democratic composition	0.484	0.403	0.000	1.000	1,748
Elected commission	0.217	0.412	0.000	1.000	1,750
Statewide Democratic share	0.493	0.096	0.218	0.748	1,750
Log population	15.114	1.014	13.025	17.492	1,750
Log GDP	12.093	1.076	9.670	15.012	1,750
Manufacturing share of GDP	0.057	0.016	0.004	0.200	1,750

Notes: State-year panel, 50 states, 1990–2024.

Composition, Selection, and Renewable Share

	Renewable Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition _{<i>t</i>-1}	-0.002 (0.013)	0.005 (0.011)	-0.000 (0.012)	0.003 (0.011)	-0.006 (0.009)
Elected commission _{<i>t</i>-1}	0.027 (0.032)	0.062 (0.047)	0.043 (0.035)	0.046 (0.035)	-0.103*** (0.021)
Composition × elected _{<i>t</i>-1}	-0.038 (0.050)	-0.028 (0.057)	-0.000 (0.042)	0.003 (0.043)	0.028 (0.019)
RPS active _{<i>t</i>-1}				-0.015 (0.014)	
Comp. effect, elected	-0.040 (0.048)	-0.023 (0.056)	-0.001 (0.040)	0.006 (0.040)	0.021 (0.018)
Observations	1,698	1,698	1,698	1,698	1,698
States	50	50	50	50	50
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

Lagged regressors; HC1 SEs clustered by state; * $p < .10$, ** $p < .05$, *** $p < .01$

Elected–Appointed Differences at Representative Compositions

State and year FE with state-specific trends

PUC Democratic composition	Difference (pp)	Clustered SE	95% CI	p-value
25%	-9.61	1.74	[-13.12, -6.11]	<0.001
50%	-8.92	1.45	[-11.83, -6.01]	<0.001
75%	-8.22	1.27	[-10.78, -5.67]	<0.001
Observations	1,698			
States	50			

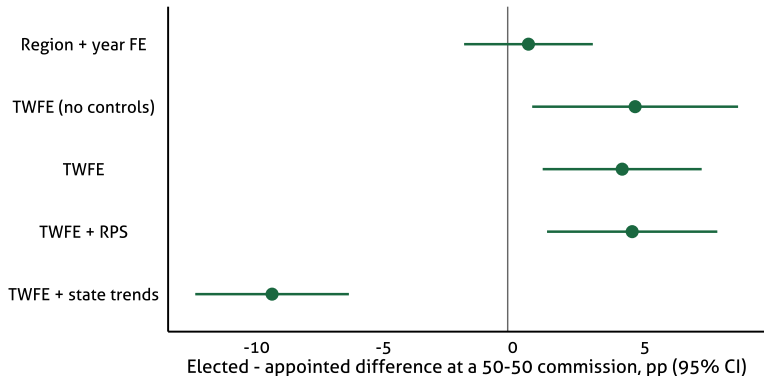
Notes: Entries are the estimated difference in the Share of Renewable Energy for Elected-minus-Appointed Commissions, expressed in percentage points, at the one-year-lagged PUC Democratic composition. Estimates are linear combinations of the elected-system coefficient and the composition-by-election interaction from the state-and-year fixed-effects model with state-specific linear trends. HC1 standard errors are clustered by state; confidence intervals and p-values use 49 cluster degrees of freedom.

Inference With Few Switching States (Full)

Specification	Difference (pp)	Cluster SE	<i>p</i> -value			Identifying states	<i>N</i>
			CR1	Wild	Placebo		
Cross-state (region and year FE)	0.8	1.2	0.519	0.557		12	1,698
+ partisanship controls	0.3	0.9	0.741	0.760		11	1,350
Cross-state, Democratic-leaning electorates	-1.5	1.0	0.156	0.143		12	845
State and year FE	4.3	1.5	0.006	0.158	0.142	2	1,698
State FE and state trends	-8.9	1.4	<0.001	0.217	0.009	2	1,698
+ partisanship controls	-4.7	0.5	<0.001	0.054		2	1,350
dropping New Mexico	-6.3	1.4	<0.001	0.407		1	1,664
dropping Tennessee	-10.3	1.0	<0.001	0.632		1	1,664

Notes: The difference is the elected-minus-appointed renewable energy share at a 50–50 Commission, in percentage points, from the specification of Table (paper) (one-year-lagged composition, selection system, and economic controls). *Identifying states:* in cross-state rows, the number of states with an elected Commission in the sample; in fixed-effects rows, the number of states whose selection system changes, which are the only source of variation in the elected indicator once state fixed effects are included. CR1: standard errors clustered by state, t with $G - 1$ degrees of freedom. Wild: wild cluster restricted bootstrap with Webb weights and 9,999 draws, null imposed. Placebo: Tennessee and New Mexico are dropped, their selection histories are assigned to every ordered pair of the remaining states, and the p -value is the share of placebo estimates at least as large in absolute value as the actual estimate. Democratic-leaning electorates: state-years with a normal-vote Democratic share above 50 percent. Rows adding partisanship control for the normal-vote Democratic share, the governor's party, and the Democratic share of state legislative seats (Snyder state-year file), and therefore cover 1990–2022 and exclude Nebraska.

The Elected–Appointed Difference Across Specifications



Elected minus appointed renewable share at a 50–50 Commission. Whiskers are 95% CIs on cluster df; SEs clustered by state.

Commission Authority and Utility Structure

Every elected Commission is in a vertically integrated state; many appointed ones are restructured

	Retail-choice ctrl		Vert. integrated		PUC covers gen		Both	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PUC Democratic composition _{t-1}	0.001 (0.011)	-0.006 (0.008)	0.012 (0.015)	-0.001 (0.011)	-0.003 (0.012)	-0.007 (0.009)	0.007 (0.015)	-0.001 (0.012)
Elected commission _{t-1}	0.036 (0.036)	-0.102*** (0.018)	0.027 (0.036)	-0.096*** (0.020)	0.043 (0.033)	-0.117*** (0.016)	0.019 (0.035)	-0.107*** (0.017)
Composition × elected _{t-1}	0.006 (0.042)	0.028 (0.018)	0.013 (0.044)	0.018 (0.021)	0.003 (0.043)	0.026 (0.020)	0.019 (0.044)	0.017 (0.022)
Retail choice active _{t-1}	-0.028** (0.013)	0.022** (0.010)						
Comp. effect, elected	0.008 (0.040)	0.022 (0.017)	0.026 (0.040)	0.017 (0.019)	-0.001 (0.041)	0.019 (0.019)	0.027 (0.040)	0.016 (0.019)
Elect effect at 50%	0.039** (0.016)	-0.088*** (0.012)	0.034** (0.016)	-0.087*** (0.012)	0.044*** (0.014)	-0.104*** (0.010)	0.028* (0.016)	-0.098*** (0.012)
Observations	1,698	1,698	1,190	1,190	1,630	1,630	1,122	1,122
States	50	50	35	35	48	48	33	33
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State trends	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Outcome is the renewable energy share. Columns (1)–(2) add an indicator for active retail-choice restructuring; columns (3)–(4) restrict to states that never restructured; columns (5)–(6) exclude Tennessee and Nebraska, whose commissions do not govern generation (TVA; all-public power); columns (7)–(8) apply both restrictions. The election effect is the elected-minus-appointed difference in the outcome at an evenly split commission. HC1-style standard errors clustered by state in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Blue States: Composition and Selection Design

	Renewable Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition _{t-1}	-0.014 (0.011)	-0.001 (0.012)	0.003 (0.011)	0.003 (0.011)	-0.004 (0.008)
Elected commission _{t-1}	-0.057** (0.024)	-0.026 (0.027)	-0.045* (0.027)	-0.046* (0.027)	-0.111*** (0.021)
Composition × elected _{t-1}	0.084** (0.036)	0.087** (0.040)	0.087*** (0.033)	0.087** (0.036)	0.037 (0.036)
RPS active _{t-1}				-0.001 (0.017)	
Marginal effect of composition, elected	0.071** (0.034)	0.086** (0.038)	0.089*** (0.031)	0.090** (0.035)	0.033 (0.035)
Observations	845	845	845	845	845
States	46	46	46	46	46
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

State-years with statewide Democratic share > 0.5. HC1 SEs clustered by state.

Composition as an Outcome

	Appointer rule (main measure)				Personal party (MSU benchmark)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Elected commission	-0.165** (0.066)	-0.080 (0.057)	0.199*** (0.058)	0.308*** (0.034)	-0.162** (0.066)	-0.076 (0.057)	0.099* (0.059)	0.223*** (0.048)
Voter Democratic share (10 pts)		0.124*** (0.027)	0.016 (0.018)	0.123*** (0.031)		0.082*** (0.028)	0.021 (0.019)	0.104*** (0.026)
Voter share $\times$ elected		0.053 (0.044)	0.152*** (0.038)	-0.015 (0.038)		0.109** (0.046)	0.164*** (0.040)	0.029 (0.037)
Democratic governor			0.607*** (0.044)				0.378*** (0.041)	
Democratic governor $\times$ elected			-0.537*** (0.083)				-0.308*** (0.077)	
Fixed effects	Year	Year	Year	State, year	Year	Year	Year	State, year
State-years	1,748	1,748	1,624	1,748	1,464	1,464	1,393	1,464
States	50	50	50	50	50	50	50	50

Notes: OLS. The outcome is the Democratic share of party-classified sitting Commissioners on July 1. Voter Democratic share is the state normal vote, centred at 50 percent and measured in 10-point units, so the coefficient on *Elected commission* in columns 2–4 and 6–8 is the elected–appointed difference in a state whose voters split evenly. Governor party is from the Snyder state-year file (1990–2022). Under the appointer rule an appointed Commissioner takes the party of the appointing authority; the personal-party benchmark uses each Commissioner's own party. Standard errors clustered by state in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

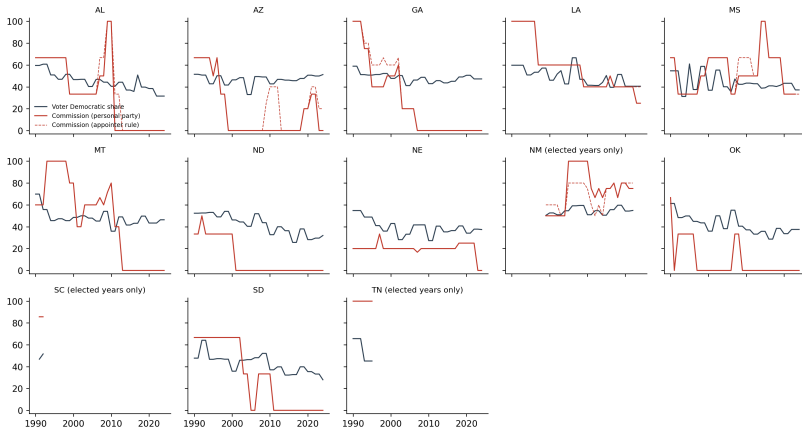
Mismatched Governors and Turnover

455 of 1,262 appointed state-years have a governor whose party runs against the state's lean

System	State lean and governor	State- years	Voter Dem. share	Commission Dem. share	
				Appointer rule	Personal party
Appointed	Democratic-leaning, Democratic governor	442	57.5	82.4	70.6
	Democratic-leaning, Republican governor	272	56.8	21.9	31.1
	Republican-leaning, Democratic governor	183	43.4	85.6	73.1
	Republican-leaning, Republican governor	365	40.9	21.4	31.8
Elected	Democratic-leaning, Democratic governor	53	55.6	67.6	67.6
	Democratic-leaning, Republican governor	58	54.7	51.1	51.4
	Republican-leaning, Democratic governor	51	43.6	29.3	29.0
	Republican-leaning, Republican governor	200	41.3	26.7	25.2
<i>Appointed commissions in mismatch years, by how long the governor's party has held office</i>					
	Democratic-leaning, Republican governor, first two years	66		48.1	48.0
	Democratic-leaning, Republican governor, third year on	206		13.5	25.6
	Republican-leaning, Democratic governor, first two years	45		70.2	64.2
	Republican-leaning, Democratic governor, third year on	138		90.7	75.7

Notes: State-years 1990–2022 with a Democratic or Republican governor. A state leans Democratic when its normal-vote Democratic share is at least 50 percent. Commission shares are means of the state-year Democratic share of party-classified sitting Commissioners on July 1; the personal-party benchmark covers fewer state-years. Governor party from the Snyder state-year file.

Composition in the Electing States



Solid red: personal-party measure; dashed red: appointer rule; dark line: voter Democratic share. NM, SC, and TN shown only for elected years.

The RPS Pathway: Stage 1

Controlling for the electorate and legislature, the composition coefficient falls from 0.18 to 0.02

	RPS active (1)	RPS active (2)	RPS active (3)	RPS active (4)	First adoption (5)
PUC Democratic composition _{t-1}	0.183** (0.074)	0.080 (0.065)	0.015 (0.071)	0.115** (0.055)	0.015 (0.023)
Elected commission _{t-1}	-0.216* (0.126)	-0.169 (0.111)	-0.145 (0.106)	0.322*** (0.114)	-0.001 (0.019)
Composition × elected _{t-1}	0.343** (0.168)	0.246* (0.148)	0.218 (0.146)	0.047 (0.127)	0.010 (0.031)
Voter Democratic share		1.627*** (0.250)	0.365 (0.310)	0.214 (0.211)	0.198*** (0.074)
Democratic governor			0.045 (0.042)	0.013 (0.040)	0.009 (0.015)
Democratic legislative seat share			0.841*** (0.159)	1.245*** (0.186)	0.152*** (0.051)
Fixed effects	Region, year	Region, year	Region, year	State, year	Region, year
State-years	1,698	1,698	1,350	1,350	908

Notes: Linear probability models. All columns include one-year-lagged log population, log real GDP, and manufacturing share. Governor party and legislative seat shares are from the Snyder state-year file (1990–2022); Nebraska's nonpartisan legislature is missing. The adoption hazard (column 5) excludes Iowa, whose RPS predates the panel, and drops states after their first adoption. Standard errors clustered by state. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

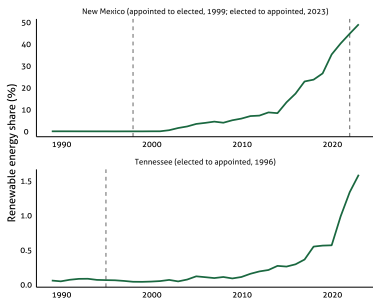
The RPS Pathway: Stage 2

Neither RPS status nor years since adoption predicts the renewable share

		Renewable energy share		
	(1)	(2)	(3)	(4)
RPS active _{<i>t</i>-1}	0.008 (0.016)	-0.013 (0.015)	-0.021 (0.022)	-0.002 (0.009)
Years since RPS adoption _{<i>t</i>-1}			0.001 (0.003)	-0.003 (0.002)
Fixed effects	Region, year	State, year	State, year	State, year, trends
State-years	1,698	1,698	1,698	1,698

Notes: The outcome is the non-hydro renewable share of generation. All columns include one-year-lagged log population, log real GDP, and manufacturing share. Standard errors clustered by state. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Synthetic Control on the Transitions



- TN 1996: -2.5 pp vs. synthetic (rank $p = 0.59$), opposite to the prediction
- NM 1999: $+6.8$ pp ($p = 0.10$), opposite to the prediction
- NM 2023: two post-years, poor pre-fit
- Pooled counterfactual estimator (Liu, Wang, Xu 2024): 0.015 (SE 0.101)

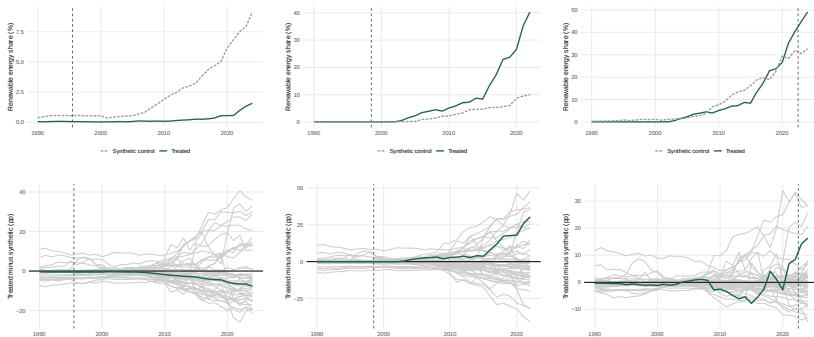
Uninformative rather than disconfirming: non-hydro renewables were near zero everywhere until the mid-2000s, so the 1990s pre-periods are flat lines any donor mix matches (NM pre-RMSPE 0.01 pp); post-1999 NM growth reflects the legislature's RPS and the national wind build-out; TN is supplied by TVA

Synthetic-Control Estimates

	Tennessee (elected to appointed, 1996)	New Mexico (appointed to elected, 1999)	New Mexico (elected to appointed, 2023)
Average post-transition gap (pp)	-2.49	6.81	15.29
Pre-transition RMSPE (pp)	0.45	0.01	3.33
Post/pre RMSPE ratio	7.4	852.9	4.6
Permutation rank p -value	0.587	0.104	0.146
Leave-one-donor-out range (pp)	[-4.32, -1.37]	[3.99, 6.80]	[15.15, 25.14]
Augmented SCM (ridge) gap (pp)	—	—	—
90% conformal interval	[—, —]	[—, —]	[—, —]
Positive-weight donors	39	7	5
Placebo units	45	47	47

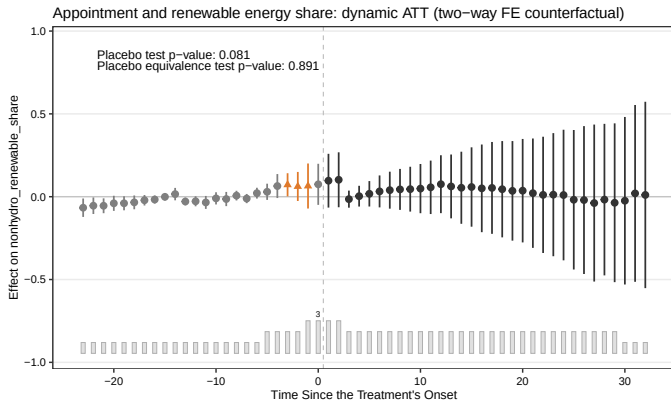
Notes: Outcome is the renewable energy share. Donors are states whose selection method never changes; weights minimize pre-transition prediction error on the outcome path and predictors. The rank p -value is the share of placebo units with a post/pre RMSPE ratio at least as large as the treated state's. The augmented estimator adds a ridge outcome model to correct for imperfect pre-period fit (Ben-Michael, Feller, and Rothstein 2021); its interval is a conformal prediction interval.

Synthetic Control: Paths and Placebos



Left to right: Tennessee 1996, New Mexico 1999, New Mexico 2023. Top: treated vs. synthetic. Bottom: treated gap (bold) against in-space placebos.

Pooled Counterfactual Estimator



Effect of appointment on the renewable share by time relative to the change, all three transitions, treatment allowed to reverse. 95% bootstrap bands.

Including Conventional Hydro

	Renewable and Hydro Energy Share				
	(1)	(2)	(3)	(4)	(5)
PUC Democratic composition _{t-1}	0.068 (0.052)	0.001 (0.012)	-0.008 (0.013)	-0.007 (0.014)	-0.009 (0.008)
Elected commission _{t-1}	0.059 (0.079)	0.040 (0.029)	0.029 (0.026)	0.030 (0.027)	-0.083*** (0.027)
Composition × elected _{t-1}	-0.122 (0.095)	0.015 (0.038)	0.024 (0.034)	0.025 (0.034)	0.034* (0.019)
RPS active _{t-1}				-0.003 (0.025)	
Comp. effect, elected	-0.054 (0.073)	0.016 (0.033)	0.016 (0.029)	0.018 (0.030)	0.024 (0.018)
Observations	1,698	1,698	1,698	1,698	1,698
States	50	50	50	50	50
State fixed effects	No	Yes	Yes	Yes	Yes
Region fixed effects	Yes	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Econ. controls	Yes	No	Yes	Yes	Yes
State trends	No	No	No	No	Yes

Outcome includes conventional hydro. HC1 SEs clustered by state.

The Commissioner Roster: Remaining Limitations

- Historical seat capacity is documented unevenly: 1,516 of 1,850 state-year capacity records are assumed (usually current size carried back); 41 states rely on an assumed capacity for at least one year
- Matters for the structural audit, not the composition measure, which uses only Commissioners recorded as serving
- 66 July 1 snapshots exceed assumed capacity; 153 overlapping-service spans remain (holdovers, contested succession dates)
- 240 state-years below the authorized maximum: vacancies, or maxima above normal staffing (New York)
- 215 spells carry a manual-review flag, mostly successor-bounded end dates
- 16 of 1,287 spells unclassified: 6 elected without party evidence (LA, MS, MT); 10 chosen by divided legislative or mixed selectors (SC, TN, VA). Costs 2 state-years of coverage

B. Voters

Energy Ballot Measures by Type, 2000–2024

Measure type	Statewide	Local	Total
Fossil fuels, facility siting, and other energy	13	204	217
Bond, tax, and finance	12	70	82
Renewable mandates and subsidies	11	64	75
Utility governance	7	0	7
Public power, transmission, and aggregation	5	1,056	1,061
Carbon pricing and climate policy	5	23	28
Nuclear	0	5	5
All energy measures	53	1,422	1,475

Notes: Energy-related measures placed before voters, 2000–2024. Statewide measures are the complete, verified list of statewide votes (53, 2000–2024). Local measures are those collected so far in the local ballot-measure dataset (v1.8; 1,422 questions, 2000–2024), which is not a national census: 989 are Illinois questions from the state’s official referenda list, most of them electricity-aggregation referenda; 95 are California questions, screened from the California Elections Data Archive’s list of every local measure in the state; and 67 are Arizona questions, most of them gas and electric franchise votes screened from Stifel Public Finance’s unofficial summaries of Arizona local elections, 2011–2025 (see appendix). Types are assigned from measure text. Groups: carbon pricing includes building electrification and gas measures; public power includes community choice aggregation; the residual group covers fossil-fuel extraction and infrastructure, facility siting, and other energy measures.

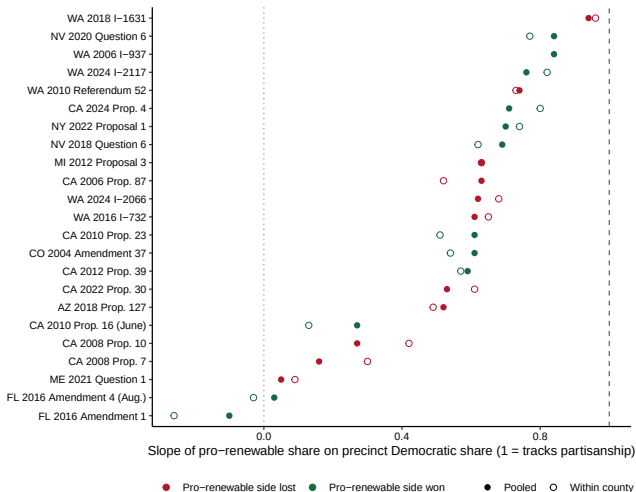
Statewide Renewable and Climate Votes, 2000–2024

State	Year	Measure and policy	Pro-renewable vote	Outcome for renewables
CO	2004	Amendment 37 — renewable energy requirement	Yes	Win
CA	2006	Proposition 87 — alternative energy funding	Yes	Loss
WA	2006	Initiative 937 — renewable energy standard	Yes	Win
CA	2008	Proposition 7 — renewable energy standards	Yes	Loss
CA	2008	Proposition 10 — alternative fuel and renewables	Yes	Loss
MO	2008	Proposition C — utility RPS mandate	Yes	Win
FL	2008	Amendment 3 — renewable energy devices excluded from property assessment	Yes	Win
ME	2010	Question 2 (primary) — renewable energy bond	Yes	Win
CA	2010	Proposition 16 (primary) — utility-backed supermajority for public power and community choice [†]	No	Win
CA	2010	Proposition 23 — suspend the state climate law (AB 32)	No	Win
WA	2010	Referendum 52 — energy-efficiency bonds for schools	Yes	Loss
MI	2012	Proposal 12-3 — 25% RPS by 2025	Yes	Loss
CA	2012	Proposition 39 — corporate tax change funding clean energy	Yes	Win
FL	2016	Amendment 1 — utility-backed restriction on rooftop solar [†]	No	Win
FL	2016	Amendment 4 (primary) — solar tax exemption	Yes	Win
WA	2016	Initiative 732 — revenue-neutral carbon tax	Yes	Loss
AZ	2018	Proposition 127 — 50% RPS by 2030	Yes	Loss
NV	2018	Question 6 — 50% RPS by 2030	Yes	Win
WA	2018	Initiative 1631 — carbon fee	Yes	Loss
NV	2020	Question 6 — 50% RPS by 2030 (required second vote)	Yes	Win
ME	2021	Question 1 — block NECEC clean-energy corridor*	No	Loss
NY	2022	Proposal 1 — environmental bond incl. climate mitigation and clean energy	Yes	Win
CA	2022	Proposition 30 — income tax for zero-emission vehicles and wildfire prevention	Yes	Loss
CA	2024	Proposition 4 — climate bond incl. clean energy	Yes	Win
WA	2024	Initiative 2066 — prohibit restrictions on natural gas [‡]	No	Loss
WA	2024	Initiative 2117 — repeal cap-and-invest program	No	Win

Statewide Votes by Precinct Partisan Lean

	Result	Pro-	Dem.	Slope	Dem. share	Pro – Dem.	Strong Dem. (>60%)		Lean Dem. (50–60%)	
		renewable	share		at 50% pro	(Dem. $\geq$ 70%)	Pro majority	Median pro	Pro majority	Median pro
CO 2004 (Amendment 37)	W	53.6	47.6	0.61	41.5	-5.1	84	68	83	57
CA 2006 (Prop. 87)	L	45.4	41.0	0.63	48.5	-13.0	86	62	64	54
WA 2006 (I-937) ^d	W	51.7	58.7	0.84	56.7	–	100	60	53	50
CA 2008 (Prop. 7)	L	35.6	62.3	0.16	>100	-42.6	8	38	3	34
CA 2008 (Prop. 10)	L	40.6	62.3	0.27	97.4	-35.8	34	46	9	40
CA 2010 (Prop. 16, June) ^c	W	54.6	53.3	0.27	36.2	-19.4	61	56	57	52
CA 2010 (Prop. 23)	W	61.5	56.8	0.61	37.9	-5.2	98	72	93	61
WA 2010 (Referendum 52)	L	46.1	52.1	0.74	57.2	-10.9	84	62	27	46
CA 2012 (Prop. 39)	W	61.1	61.9	0.59	42.9	-8.9	99	70	93	57
MI 2012 (Proposal 3)	L	37.7	54.8	0.63	74.0	-28.7	66	56	1	37
FL 2016 (Amendment 1) ^e	W	49.0	49.4	-0.10	–	-40.2	26	44	53	51
FL 2016 (Amendment 4, Aug.) ^c	W	72.6	44.3	0.03	–	-12.2	98	74	98	76
WA 2016 (I-732)	L	40.8	58.8	0.61	73.5	-26.6	48	50	1	38
AZ 2018 (Prop. 127)	L	32.1	52.1	0.52	86.6	-35.5	21	39	0	35
NV 2018 (Question 6)	W	59.3	52.2	0.69	38.6	-3.9	98	71	96	64
WA 2018 (I-1631)	L	43.5	58.5	0.94	65.4	-15.1	67	56	1	39
NV 2020 (Question 6)	W	57.9	51.2	0.84	41.5	-1.7	100	71	97	63
ME 2021 (Question 1) ^d	L	40.8	54.5	0.05	–	-36.5	6	40	6	40
CA 2022 (Prop. 30)	L	42.4	59.2	0.53	73.5	-26.9	47	50	9	41
NY 2022 (Proposal 1) ^b	W	76.6	62.8	0.70	26.1	5.9	100	92	100	72
CA 2024 (Prop. 4)	W	59.8	60.4	0.71	46.7	-6.4	98	69	82	58
WA 2024 (I-2066)	L	48.3	59.5	0.62	62.3	-17.3	72	54	15	45
WA 2024 (I-2117)	W	62.0	59.5	0.76	43.9	-1.7	100	69	96	59

Slopes on Precinct Partisanship



Slope of pro-renewable share on Democratic share, pooled (filled) and within county (hollow). One = support tracks partisanship exactly.

Cost-Imposing vs. Advisory Local Measures (Full)

	Mean pro-renewable share (N)		Difference	Permutation	Pro side won, % (N)	
	Cost-imposing	Advisory	[95% CI]	<i>p</i>	Cost-imposing	Advisory
<i>A. Differences in means</i>						
All places	56.6 (41)	68.7 (47)	-12.1 [-19.1, -4.8]	0.003	70 (53)	77 (81)
Democratic share ≥ 60%	58.6 (36)	76.6 (37)	-18.0 [-23.0, -13.1]	0.000	75 (44)	92 (53)
Democratic share ≥ 70%	58.5 (30)	80.4 (23)	-21.9 [-27.1, -16.7]	0.000	71 (34)	97 (39)
Within the frame (F1, F2, F2b)	52.2 (21)	75.7 (39)	-23.5 [-30.4, -16.6]	0.000	61 (28)	84 (73)
Core tier (b): drops minor-share bundles	57.8 (35)	68.7 (47)	-10.8 [-18.3, -3.2]	0.010	76 (45)	77 (81)
Core tier (c): explicit, primary purpose	58.4 (21)	68.7 (47)	-10.3 [-19.3, -1.6]	0.051	73 (22)	77 (81)
Tier (b), Democratic share ≥ 60%	59.3 (33)	76.6 (37)	-17.4 [-22.8, -12.0]	0.000	80 (40)	92 (53)
Advisory excluding local siting votes	56.6 (41)	76.3 (38)	-19.7 [-25.1, -14.1]	0.000	70 (53)	87 (62)
and Democratic share ≥ 60%	58.6 (36)	77.4 (36)	-18.8 [-23.8, -13.7]	0.000	75 (44)	94 (51)
<i>B. Regression of pro-renewable share on cost-imposing indicator: coefficient (robust SE)</i>						
Bivariate	-12.1 (3.8)		N = 88			
Jurisdiction type and year	9.6 (8.4)		N = 88			
Jurisdiction fixed effects	-32.3 (6.8)		N = 13; 2 jurisdictions (Berkeley, San Francisco)			
Dem. share ≥ 60%, with Dem. share and year	-18.9 (2.6)		N = 73			
No siting advisories, jurisdiction type and year	-16.5 (6.7)		N = 79			

◀ Back

Local Measures: Descriptives and Frame

	Measures	With returns	Pro side won, % (N)		Mean pro share	Median		
<i>A. By cost class</i>								
Cost-imposing (core set)	53	41	70 (53)		56.6	58.0		
Direct cost (tax, fee, bond)	38	28	84 (38)		60.5	61.0		
Compliance mandate	5	4	40 (5)		44.0	49.3		
Public ownership	10	9	30 (10)		49.9	48.3		
Advisory (non-binding)	85	47	77 (81)		68.7	76.7		
Binding, no new cost	16	9	86 (14)		63.1	57.6		
	Cost-imposing (core)				Advisory (no cost)			
	N	Returns	Won %	Mean	N	Returns	Won %	Mean
<i>B. By frame part</i>								
F1: Illinois census (1995–2026)	1	0	100	–	39	1	74	42.7
F1: California census (1996–2025)	13	13	31	51.6	3	3	100	82.4
F1: Massachusetts district questions	0	0	–	–	35	35	91	76.1
F2: Ballotpedia coverage units	9	5	78	49.2	0	0	–	–
F2b: same units, pre-coverage years	5	3	100	60.0	0	0	–	–
Outside the frame	25	20	80	61.1	8	8	12	34.5
<i>C. By period</i>								
2000–2009	15	13	60	54.0	20	11	90	81.5
2010–2017	14	10	79	58.6	24	9	50	35.8
2018–2026	24	18	71	57.3	41	27	86	74.4

Local Measures: Precinct Benchmarks

Measure	Cost	Precincts	Pro %	Dem %	Slope on Dem % (SE)	Pro – Dem, Dem $\geq 70\%$ (N)	Same-ballot tax Yes %	Pro – tax
<i>A. Cases 2018–2026</i>								
Portland Measure 26-201 2018	direct	81	65.3	81.5	0.72 (0.04)	-16.7 (52)	69.0	-3.7
San Francisco Proposition A 2018	direct	597	77.2	90.9	1.34 (0.03)	-13.8 (597)	55.5	21.7
Berkeley Measure HH 2020	direct	33	47.0	95.9	0.28 (0.74)	-47.2 (33)	–	–
Denver Ballot Measure 2A 2020	direct	356	62.3	81.4	0.96 (0.03)	-19.7 (325)	–	–
Portland Question C 2020	mandate	11	49.8	67.2	0.81 (0.14)	-20.0 (5)	–	–
Ann Arbor Proposal 1 2022	direct	53	71.1	90.3	2.16 (0.29)	-16.7 (53)	63.1	7.9
Berkeley Measure L 2022	direct	31	59.4	94.7	-1.28 (1.20)	-31.0 (30)	64.9	-5.5
Boulder Ballot Issue 2A 2022	direct	58	70.5	90.7	0.72 (0.32)	-20.2 (58)	73.3	-2.9
Boulder Ballot Issue 2B 2022	direct	58	68.4	90.7	0.70 (0.32)	-22.2 (58)	73.6	-5.2
El Paso Proposition C 2022	direct	172	50.6	64.7	0.58 (0.02)	-18.2 (74)	57.1	-6.5
Burlington Question 2 2023	direct	8	67.3	–	–	–	–	–
Berkeley Measure GG 2024	direct	32	31.0	94.6	-2.84 (0.51)	-59.4 (30)	66.9	-35.9
Ipswich Question 6 2024	direct	4	44.3	65.5	–	-22.7 (1)	–	–
Portland Measure 26-253 2024	mandate	79	49.1	87.3	0.44 (0.04)	-39.1 (64)	–	–
Salt Lake City School District bond 2024	direct	140	61.1	78.3	0.39 (0.05)	-19.4 (110)	73.6	-12.5
<i>B. Earlier cases, 2001–2017 (supporting)</i>								
San Francisco Proposition B 2001	direct	648	72.6	82.0	0.91 (0.04)	-10.1 (477)	72.1	0.4
San Francisco Proposition F 2001	ownership	648	49.8	82.0	1.30 (0.04)	-31.0 (477)	72.1	-22.3
San Francisco Proposition H 2001	direct	648	53.9	82.0	1.08 (0.05)	-28.5 (477)	72.1	-18.2
San Francisco Proposition I 2001	ownership	648	48.3	82.0	1.28 (0.04)	-32.5 (477)	72.1	-23.9
San Francisco Proposition D 2002	ownership	631	46.3	81.6	1.24 (0.03)	-35.1 (550)	56.2	-9.9
Boulder County Ballot Issue 1A 2008	direct	234	63.7	72.9	0.83 (0.02)	-10.8 (142)	67.4	-3.7
San Francisco Proposition H 2008	ownership	579	38.6	86.0	1.45 (0.04)	-47.4 (570)	73.7	-35.0
Boulder Issue 2B 2011	direct	88	50.4	85.1	1.57 (0.11)	-33.2 (87)	74.9	-24.5
Boulder Question 2C 2011	ownership	88	51.9	85.1	1.58 (0.13)	-30.8 (87)	62.5	-10.6
Boulder Ballot Issue 2A 2012	direct	86	82.3	83.5	0.34 (0.11)	-1.5 (84)	85.0	-2.7
Boulder Question 2E 2013	ownership	87	66.5	86.2	1.17 (0.06)	-19.6 (86)	65.8	0.7
Boulder Question 310 2013	ownership	87	68.7	86.2	1.25 (0.08)	-16.4 (86)	65.8	2.8
Lafayette Issue 301 2013	direct	17	50.4	71.1	0.75 (0.09)	-22.0 (11)	17.4	33.0
Lafayette Ballot Issue No. 2D 2014	direct	17	45.8	71.3	0.56 (0.11)	-27.4 (11)	60.7	-14.9
Boulder Ballot Question 2P 2015	direct	87	77.2	86.2	0.98 (0.06)	-7.5 (86)	57.6	19.6
Boulder Issue 2L 2017	direct	87	51.7	86.2	1.23 (0.15)	-31.3 (86)	80.0	-28.3
<i>C. Pooled, measure fixed effects (SE clustered by measure)</i>								
2018–2026 cases		1702			0.72 (0.11)	-19.6 (2.8)		
All cases, 2001–2026		5944			0.93 (0.10)	-26.2 (3.4)		

D. Ballot-level benchmark, 2020 cast vote records

	This measure			Other tax measures, same ballot		
	Dem. Yes	Rep. Yes	APG	Dem. Yes	Rep. Yes	N
Denver 2A (climate sales tax)	75.5	12.4	63	81.8	29.1	3
Berkeley HH (utility tax)	49.4	5.3	44	75.9	23.4	5
Columbus Issue 1 (no-cost aggregation)	89.7	38.8	51	69.5	43.8	5
National local tax/spend benchmark, 2020	71.8	52.4	20			

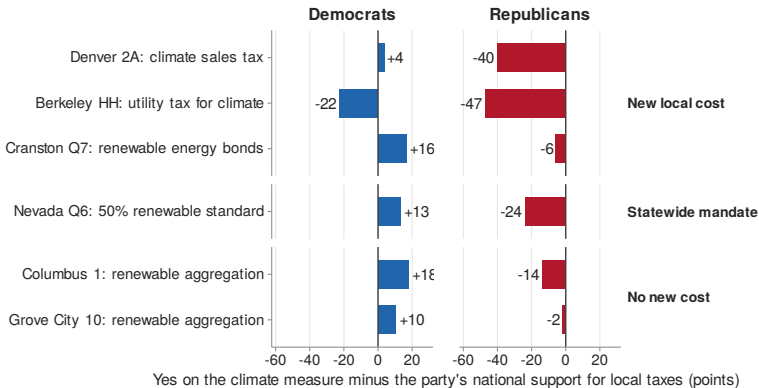
Local Measures: How the Frame Was Built

- ◉ **F1:** official lists of every local question: Illinois (1995–2026), California Elections Data Archive (1996–2025), Massachusetts
- ◉ **F2:** Ballotpedia's complete coverage: 100 largest cities from 2018, state capitals from 2021, all of California from 2016, NC 2019, VA 2025–26
- ◉ **F2b:** the same cities back to 2000 in official archives, costly measures only
- ◉ Screens: 12,149 IL referenda; 6,243 Ballotpedia measures; 11,501 CA measures; audits of 250 rejects bound misses at 1.2–2.2%
- ◉ Topic tags alone would miss half the costly measures (8 of 16)
- ◉ Cost coded from official (33 of 53) or secondary (14) ballot text, before outcomes
- ◉ Tiers: (b) drops minor-share bundles; (c) explicit, primary climate purpose

Cast Vote Records, 2020 (Full)

	Yes vote (%)			Party gap (Dem. – Rep.)	vs. party's national local tax support		vs. same voters' other local taxes on ballot	
	Dem.	Rep.	Split		Dem.	Rep.	Dem.	Rep.
<i>A. Local measures that impose a new cost</i>								
Denver, CO, Measure 2A (climate sales tax)	75.5	12.4	40.9	63.1	+3.7	-40.0	-6.1	-16.4
Berkeley, CA, Measure HH (utility users tax for climate)	49.4	5.3	16.2	44.1	-22.4	-47.1	-25.8	-17.7
Cranston, RI, Question 7 (\$5M renewable energy bonds)	88.3	46.5	66.6	41.8	+16.5	-5.9	+4.5	-19.2
<i>B. Statewide renewable mandate</i>								
Nevada Question 6 (50% renewable portfolio standard)	85.0	28.6	63.7	56.4	+13.2	-23.8	–	–
<i>C. Local measures with no new cost</i>								
Columbus, OH, Issue 1 (renewable electricity aggregation)	89.7	38.8	73.2	50.9	+17.9	-13.6	+5.4	-5.6
Grove City, OH, Issue 10 (renewable electricity aggregation)	82.2	50.2	77.0	32.0	+10.4	-2.2	+7.1	-0.3

2020 Cast Vote Records vs. the Party's National Support for Local Taxes



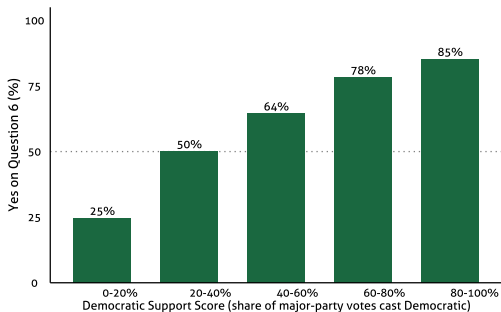
Yes vote on each measure minus the party's national 2020 support for local tax and spending measures (Dem. 71.8%, Rep. 52.4%).

California Cast Vote Records (Full)

County, year	Climate measure (paired measure)	Yes vote (%)			Party gap (Dem. – Rep.)	vs. same voters' vote on paired measure			Ballots
		Dem.	Rep.	Split		Dem.	Rep.	Split	
Marin, 2024	Prop. 4, climate bond (\$10B) (Prop. 2, school bond (\$10B))	82.4	23.0	54.0	59.4	+7.5	-10.5	+6.3	139,189
Santa Barbara, 2024	Prop. 4, climate bond (\$10B) (Prop. 2, school bond (\$10B))	84.2	20.6	55.1	63.6	+4.7	-6.0	+3.9	187,876
Marin, 2022	Prop. 30, income tax for zero-emission vehicles (Prop. 28, arts and music funding)	57.1	13.0	32.9	44.1	-27.4	-16.1	-24.0	114,402

Notes: Cast vote records from the county election offices (Dominion exports). Voters are classified by the party of their U.S. House vote; split-ticket voters are those whose House vote and top-of-ticket vote (Governor in 2022, President in 2024) went to different major parties, and the top-of-ticket vote classifies voters without a major-party House vote. The comparison columns take each voter's vote on the climate measure (Yes = 1) minus the same voter's vote on the paired measure on the same ballot, averaged within party among voters who voted on both, in percentage points. Marin and Santa Barbara are the California counties whose 2022 or 2024 records available to me link every card of a ballot; both lean Democratic, so the figures describe these electorates rather than the state.

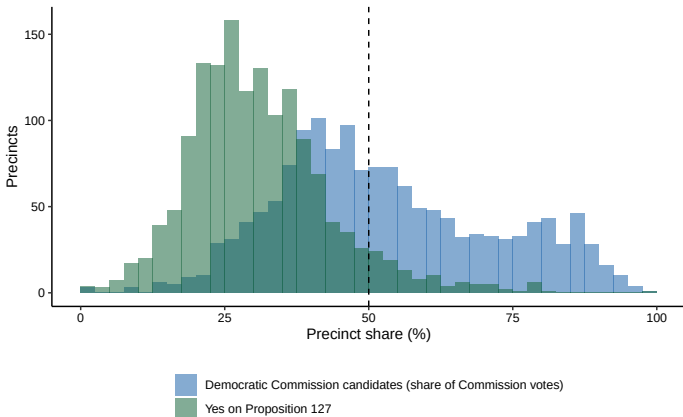
Nevada Question 6 by Democratic Support Score



Yes share on Question 6 (2020) by the ballot's Democratic Support Score: share of its partisan contests won by the Democrat.

- 85% among straight-ticket Democrats, 24% among straight-ticket Republicans
- Passed twice (2018, 2020) under conditions unique to Nevada: a credible economic case from solar resources, and no serious organized opposition

Arizona 2018: Both Distributions

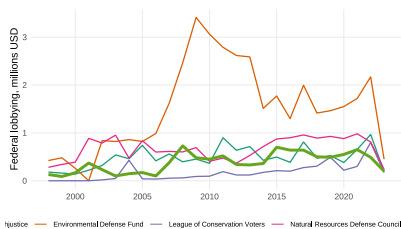
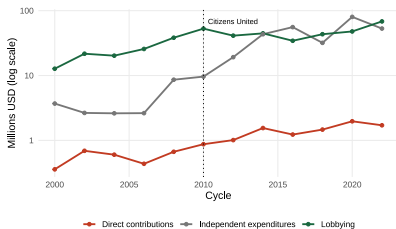


Precinct support for the Democratic Commission candidates and for Proposition 127, Arizona 2018.

C. Interest Groups

Federal Lobbying and Campaign Finance, 2000–2022

Coalition	Lobbying dollars per dollar of	
	direct contributions	direct contributions and independent expenditures
Environmental organizations	35.9	1.4
Fossil fuel producers and utilities	23.5	23.4



Environmental-sector federal spending by channel (log scale)

Federal lobbying by focal NGOs; Sierra Club in bold

With independent expenditures, greens match industry after 2014: episodic mega-donor spending in a few salient federal races, which does not scale down to a Commission race. OpenSecrets JE300; industry codes exclude nuclear, renewables, water

Georgia PSC: Candidate Content

	Candidacies
Total candidacies coded	65
No identifiable position	30
Lean pro-renewable	18
Strongly pro-renewable	9
Mixed or conflicting	4
Lean or strongly anti-renewable	4
League of Conservation Voters endorsement	0
Sierra Club endorsement	3

Notes: Hand coded Georgia PSC candidacies from 1998–2022 based on campaign coverage, debate records, ballot guides, and endorsement archives; each coding carries a documented evidence summary and sources. Categories are mutually exclusive.

2024 reviews of races in all eight electing states: platforms organized around reliability, affordability, and procedure; renewable energy largely absent

Allocation Across Channels: Tables

Resource allocation across channels

Sample	Window	Inside / direct	Inside / comparison
Federal: Green	2000–2022 cycles	35.90	1.38
Federal: Incumbents	2000–2022 cycles	23.49	23.45
State: green	2017–2024	—	38.42
State: incumbent	2017–2024	—	2.34

Organizational resources by channel, per organization–state–year

Coalition	Campaign \$	Campaign budget %	Lobbying \$	Lobbying budget %
Green	4,045	0.105	155,388	4.051
Incumbent	104,537	2.165	244,180	5.065
Adjusted green contrast	-3.268 (0.019)	-0.021 (0.000)	-0.506 (0.058)	-0.010 (0.002)

[◀ Back](#)

Selection Design and Time Allocation

Relative resources and selection design

System / contrast	Channel	Dollar ratio / log contrast	Budget ratio / fraction contrast
Appointed	lobbying	0.737	0.929
Appointed	campaign	0.038	0.049
Elected	lobbying	0.283	0.349
Elected	campaign	0.040	0.048
Green $\times$ elected	lobbying	-0.947 (0.032)	-0.030 (0.002)
Green $\times$ elected	campaign	0.056 (0.044)	0.000 (0.001)

Time allocation among NGO participants (189 organization–state–years)

Activity	Mean hours	Org-state-years
Docket preparation	641.5	189
Other inside activity	55.4	189
Total inside activity	696.9	189
Campaign labor	20.0	189
Inside / campaign mean-hours ratio	34.91	

Docket Samples and Participation

Matched pairs

Pair	Elected	Appointed	Region
1	AZ	NV	West
2	GA	KY	South
3	LA	AR	South
4	MS	WV	South

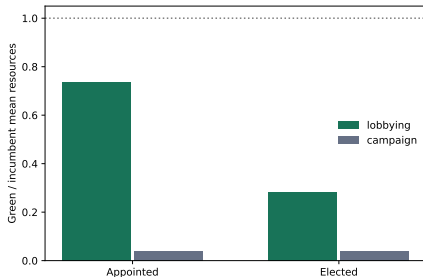
Participation and filings, first 180 days

Actor class	Participation	Filings	Substantive	Priced
Consumer nonprofits	36.9%	918	61.1%	92.9%
Fossil-fuel firms	27.5%	675	62.8%	93.6%
Environmental NGOs	91.6%	5,210	69.3%	92.8%
Other nonprofits	0.0%	0	nan%	nan%
Public advocates	74.1%	1,778	62.2%	93.3%
Renewable firms	28.9%	710	61.3%	92.8%
Utilities	97.4%	3,548	61.6%	93.1%

Organizational Practices and Competitiveness

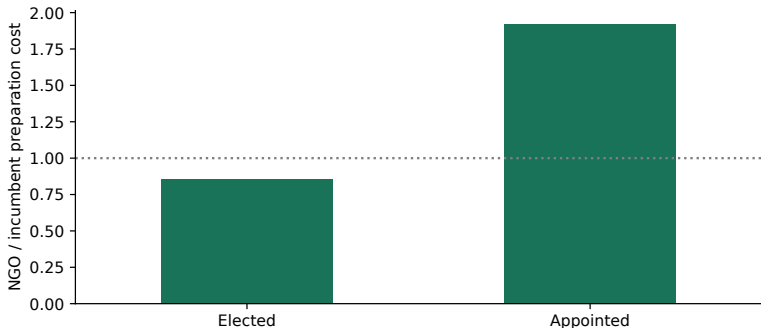
Filings in appointed systems,
first 180 days

Actor	Substantive	Testimony	Joint
Fossil-fuel firms	63.5%	21.9%	0.0%
Environmental NGOs	73.4%	27.2%	17.4%
Utilities	62.3%	22.3%	0.0%



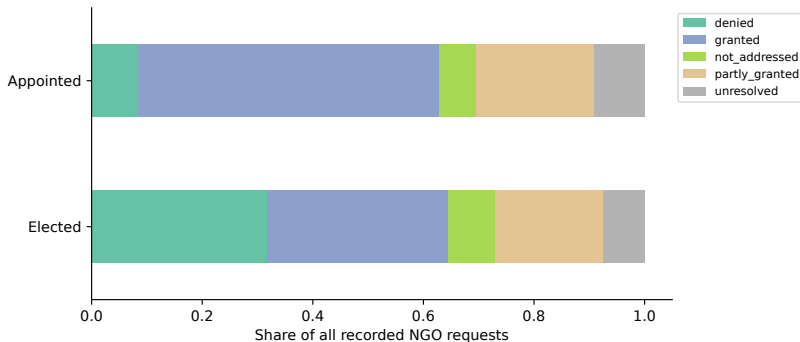
Green-to-incumbent ratios of mean resources;
one = equal measured resources (not an
equivalence test)

Preparation Resources Relative to Incumbents



NGO preparation resources relative to incumbent industries within eligible docket cohorts. The reference line marks equal aggregate measured resources, not equal access or influence.

Disposition of All Recorded NGO Requests



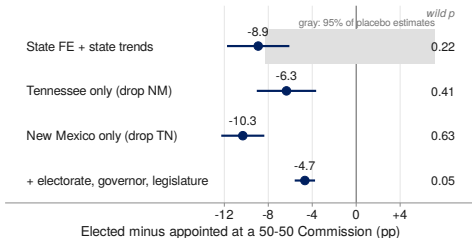
Contrast	Estimate	95% interval
Appointed minus elected	0.333	[0.266, 0.399]

Coding Requests and Final Orders

- ⦿ One row per organization, docket, and distinct requested action, with page or paragraph location in a dated filing
- ⦿ Disposition located in the final order: granted, partly granted, denied, not addressed, unresolved, withdrawn, unclear
- ⦿ Partial grants kept separate; unresolved and unaddressed requests are *not* defeats
- ⦿ Requests averaged within organization–docket, so long briefs do not add independent cases
- ⦿ The order date sets the regime; transition-spanning proceedings flagged; effort must precede the order
- ⦿ Joint filings attributed once (lead author or largest documented contribution; equal shares as sensitivity)
- ⦿ Second coder on a stratified subset

D. Anticipated Questions

Two Switching States: Strength and Limits



- **Why within-state:** elected states are windier, more rural, more Republican, so comparing across states mixes selection with geography
- **What the two cases add:** the same state under both systems; the switches run in opposite directions (TN to appointment; NM to election and back); each state alone gives a negative gap (-6.3 , -10.3) once its own trajectory is modeled
- **Where it stops:** two clusters, so wild-bootstrap $p = 0.05$ – 0.63 ; the estimate sits just outside 95% of placebo estimates (placebo $p = 0.009$)
- $\Rightarrow$ Read as two careful case studies, and test the mechanisms directly (Parts II, III)

Discounting and the Social Cost of Carbon

- ◉ **Discount-rate debate:** the weight given to future damages drives the social cost of carbon; a low rate implies aggressive action now (Stern 2007), market-based rates a gradual ramp (Nordhaus 2007; Weitzman 2007)
- ◉ **Present bias:** quasi-hyperbolic discounting shrinks delayed benefits and makes choices time-inconsistent (Laibson 1997); citizens discount policy benefits that arrive later (Jacobs and Matthews 2012)
- ◉ **Energy-efficiency gap:** consumers pass up efficiency investments that pay for themselves, at high implicit discount rates (Hausman 1979; Jaffe and Stavins 1994; Allcott and Greenstone 2012)
- ◉ **Politics of the long term:** voters reward disaster relief, not preparedness (Healy and Malhotra 2009); long-term investment must overcome electoral short-termism (Jacobs 2011)

→ The gap between voters and the welfare benchmark is the same question: whose preferences count, over what horizon

Aren't Voters Just Discounting the Future Rationally?

- ◉ Possibly, and I take revealed preferences on ballot measures as sincere
- ◉ My benchmark is social welfare, which includes the externality and future consumers; voters' own choices need not include either
- ◉ The gap may be a genuine distributional conflict between today's energy consumers and future generations, the same question as the discount-rate debate (Stern 2007; Nordhaus 2007)
- ◉ The empirical claim does not depend on resolving it: whatever its source, the electoral signal elected Commissioners face runs against acting on the transition

Why Prefer the Trend-Inclusive Estimate?

- ⊙ State FE absorb each state's time-invariant endowment; the remaining difference between specifications is the *trajectory* of development
- ⊙ Without trends, the Plains wind build-out (production tax credit, resource quality) coincides with the elected states and is attributed to the institution
- ⊙ With trends, the difference is identified from deviations around each state's own path
- ⊙ Trade-off: trends also absorb an institutional effect that works *gradually*, through the slow accumulation of resource-plan and rate-case decisions, so they may understate it

Isn't the Appointer Rule Mechanical?

- ◉ It is partly tied to the governor by construction, so every composition result is also shown with each Commissioner's own party (MSU archive)
- ◉ On personal party, appointed Commissions still follow the governor: a Democratic governor is associated with a Commission 38 pts more Democratic (61 on the appointer rule)
- ◉ Mismatch cases: appointed Commissions in blue states with Republican governors are as Republican as those in red states (31% vs. 32% Democratic)
- ◉ Disagreement with personal party (72% agreement for appointed seats) is plausibly closer to classical error → attenuation
- ◉ Where the two measures disagree (the blue-state gradient), I say so: E4 is not established

Is It Market Structure or Generation Authority?

- ⊙ Every elected Commission sits in a vertically integrated state; many appointed ones are restructured
- ⊙ The mechanism requires the gap where the Commission governs generation: IRPs, certificates, rate-basing
- ⊙ Elected–appointed difference at 50–50, state FE and trends:
 - ⊙ controlling for retail choice: -8.8 pp
 - ⊙ never-restructured states only (35): -8.7
 - ⊙ excluding TN (TVA) and NE (public power): -10.4
 - ⊙ both restrictions: -9.8
- ⊙ Same caveat as the main estimate: identified from the switching states

Are Ballot Measures a Selected Sample?

- ◉ Yes, and the selection likely *inflates* support: measures need organized backing to qualify, and initiative states are disproportionately Western and track majority opinion closely (Matsusaka 2004; Lupia and Matsusaka 2004)
- ◉ Statewide list is complete for 2000–2024; local completeness is claimed only within a frame fixed before looking at outcomes
- ◉ Costly and advisory sets are defined by question content, not outcome: a costly measure that passed counts as much as one that failed
- ◉ Places differ: the cleanest comparison is within Berkeley and San Francisco (–32 pts) and against same-ballot taxes
- ◉ CVR counties lean Democratic, so they describe those electorates, not the state

Does Federal Behavior Tell Us About the States?

- ◉ No comprehensive disclosure of state-level lobbying exists; the federal record is the one place both channels are fully measured for the same actors
- ◉ The organizations are the same at both levels: Sierra Club, NRDC, EDF, LCV operate in Washington and in state capitals
- ◉ The allocation reveals the technology of influence they have built, which travels with the organization
- ◉ The state evidence (Commissioner races, ballot committees, dockets, NM Amendment 1) then asks whether they behave the same way where the stakes are Commissioners
- ◉ Small campaign contributions are not evidence of small influence (Ansolabehere et al. 2003), which is why I measure inside effort directly

Isn't New Mexico a Single, Bundled Case?

- ◉ Yes: the 2023 change bundled selection method, membership, and size ($5 \rightarrow 3$), alongside other policy and economic developments
- ◉ Amendment 1 is revealed-preference evidence, not an estimate of an effect: the same organizations that decline to fund Commissioner candidates paid to end Commissioner elections
- ◉ In the docket analysis: 2020–2022 anticipation kept separate; transition-spanning proceedings flagged and excluded as a sensitivity check; utilities as a comparison series, not a placebo
- ◉ Tennessee also switched (1996), but its proceeding records before 2000 could not be obtained

Why Would Pro-Consumer Groups Prefer Less Democracy?

- ◉ Not less democracy in general: an arena where their resources (expertise, testimony, persistence) count, and where incumbents' electoral advantages (money, votes, a simple message) sit idle
- ◉ Under appointment the governor builds the coalition for them; they need not assemble an electoral majority
- ◉ Consistent with Fournaies and Fowler (2022) on insurance regulation: influence runs largely through channels other than transfers to officeholders
- ◉ Normative question left open: rational adaptation to US institutions, or a departure from a grassroots mandate? What does it mean for the legitimacy of appointed regulation?

E. My Argument in Detail

My Argument in Brief

Within this new interest group structure, **electoral accountability gives rise to regulators who are less likely to promote a long-term energy transition**, and more likely to be staunchly pro-incumbent industry

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- ◉ Green interests are overshadowed in elections: incumbents' decades of presence, status-quo bias, a simpler message; few voters pay more now for renewables
- ◉ More important: green interests are **strong inside the beltway**, and have adapted to prioritize lobbying over campaigns

→ Lower renewable growth on average where voters elect the Commission, varying with the electorate's partisanship, preference distributions, and resources

How Regulator Selection Affects Alternative Energy: The Electoral Arena

Elections are themselves arenas in which organized interests can dominate diffuse voters (Stigler 1971). The pro-transition coalition is weak on both flanks

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- ◉ **Business flank outsized:** **alternative energy firms** vs. **fossil fuel producers**, among the most politically organized industries, and **monopoly utilities**, who campaign at a scale entrants cannot match
 - ◉ Incumbents also command votes, through the workforces and communities that depend on them (Bombardini and Trebbi 2020)
- ◉ **Consumer flank small:** the largest bloc is status-quo consumers who care about price, and their message aligns with the more powerful business interest
 - ◉ Fossil fuel firms' long-lived political apparatus cemented economic identities that leave consumers skeptical of renewables (Dewitte 2026)
 - ◉ Voters willing to pay more for renewables are a small subset, even in the most Democratic states

The Electoral Arena: A Hard Message to Carry

- Benefits of the transition are complex: externalities and short-versus long-term trade-offs, with no ready heuristics (Lupia 1994)
- Infrastructure adds visible short-term costs, and projects disturb communities (offshore wind at Martha's Vineyard; siting backlash: Stokes 2016)
- 'Lower prices now' is simple, backed by the dominant business interest, reinforced by status-quo bias
- Elected commissioners act as if punished for price increases (Parinandi and Hitt 2018)
 - AZ Democrats won on renewables and rates in 2018, lost 2022 and 2024; GA's 2025 Democratic PSC wins ran on rates and Vogtle; Davante Lewis (LA 2022) the exception
- A natural majority that never is one at the ballot box (Przeworski and Sprague 1986)
- Off-cycle, down-ballot races favor narrow interests (Anzia 2011); initiatives return groups to an electorate with modest willingness to pay (Ansolabehere and Konisky 2014)

Regulator Selection Under Industrial Change: Inside the Beltway

Electoral accountability tilts regulator behavior toward status quo (fossil + utility interests) → slower integration of alternative energy

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What about appointments? As in Besley and Coate (2003), regulation is bundled with other issues, including alternative energy integration

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Key difference: alternative energy firms and pro-alternative energy NGOs have more influence **INSIDE THE BELTWAY**

- Influence depends on *relative* efficiency at producing pressure; arenas reward different technologies of pressure (Becker 1983)
- The task inside: show regulators integration is best for long-term pricing and stability
- Green groups compete with what they hold: expertise, testimony, persistence in resource-plan and rate cases, regulators' technocratic language
- The governor assembles the coalition for them: green voters bundled with monied allies (renewable developers, in-state manufacturers)

Inside the Beltway: Strategic Allocation

- ⊙ Appointed regulators maximize across organized interests' information and support; green NGOs punch above their electoral weight → **more balanced regulators**
- ⊙ Where fossil fuel interests dominate elections unopposed, regulators are more likely staunchly pro-fossil fuel
- ⊙ Green organizations should lobby, intervene, and cultivate appointers, and spend on elections sparingly and episodically
- ⊙ It is the *return to inside effort* under each institution; inside activity need not center on appointments
- ⊙ Consistent with insurance regulation: influence runs largely through channels other than transfers to officeholders (Fouirnaies and Fowler 2022)

Measurement implication: small campaign contributions are not evidence of small influence (Ansolabehere et al. 2003) → measure effort inside the regulatory process directly